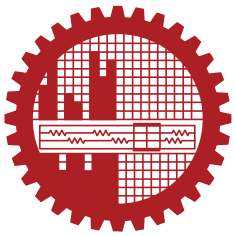
Design And Performance Analysis Of A Low-Energy Topological Field Effect Transistor Based On The Phase Transition In 1T′-WTe_2 Nanoribbons

by

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MASTER OF SCIENCE
IN
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The thesis titled, **"Design And Performance Analysis Of A Low-Energy Topological Field Effect Transistor Based On The Phase Transition In 1T′-WTe_2 Nanoribbons"**, submitted by **Joy Sharma**, Roll No.: 0422062303, Session April 2022, has been accepted as satisfactory in partial fulfillment of the requirements for the degree of Master of Science in Electrical and Electronic Engineering on 20 December 2025.

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This is to certify that the work presented in this thesis entitled, "Design And Performance Analysis Of A Low-Energy Topological Field Effect Transistor Based On The Phase Transition In 1T′-WTe_2 Nanoribbons", is the outcome of the research carried out by Joy Sharma under the supervision of Dr. Mahbub Alam, Professor, Department of Electrical and Electronic Engineering, Bangladesh University of Engineering and Technology (BUET), Dhaka-1205, Bangladesh.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

Signature of the Candidate

Joy Sharma
0422062303

Dedication

I like to dedicate my work to my parents for their unconditional love and endless support throughout my life.

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ABSTRACT

In recent years, there has been significant interest in topological materials, motivated by their unique electronic properties that offer potential for topological field-effect transistors (TFET). A two-dimensional (2D) topological insulator (TI) has quantum spin hall effect (QSH). 2D TI is defined by an insulating bulk and a conductive edge state, wherein carriers with distinct spins flow in opposite edges. The sole scattering pathway for the edge current is backscattering, which is forbidden by time reversal symmetry. Owing to their backscattering-prohibited edge states, quantum spin Hall (QSH) insulators can be used as channel materials for TFET applications. These edge states are highly robust against nonmagnetic impurities, resulting in low-energy dissipation. 1T′-WTe_2 stands as one of the most promising topological materials, because of a sizeable 0.114 eV band gap, inherent structural stability, and large spin orbit coupling (SOC). Non-trivial topologically protected edge states are susceptible to the applied vertical electric field in the nanoribbons which results in changing phase transition of the materials from topological to trivial insulator. In this work, We utilize density functional theory (DFT) together with the non-equilibrium Green's function (NEGF) formalism to analyze the electrical and transport properties of the device. Our investigation indicates a topological phase transition in 1T′-WTe_2 after applying a critical electric field, E=0.1431 V/Å , signifying the transition from the QSH phase to a trivial insulating phase. We systematically analyze the effects of channel length variation, and vacancy defects on the device's transfer characteristics. Our results demonstrate that expanding the channel length significantly suppresses the tunneling leakage current during the off state, resulting to an improvement in the subthreshold swing. We also assess crucial performance metrics—such as the on/off ratio, capacitance, delay time, and energy dissipation. This work underscores the importance of topological aspects in device performance and provides practical insights for constructing next generation, energy-efficient nanoelectronic devices.

CHAPTER 1: INTRODUCTION

1.1 Problem Statement

This era is characterized by the Internet, social networking, and mobile devices. There were approximately 5.56 billion internet users worldwide, or 67.9% of the global population, by the first quarter of 2025 [3]. The increase in internet users is remarkable, with recent data indicating that over 136 million users were added to the global connected population in the 12 months preceding October 2025. As of October 2025, the total number of active social media user identities reached 5.66 billion, with an accelerated rate of 9.6 new users every second. Extensive cloud computing networks and data centers manage the substantial volume of traffic linked to these operations. Furthermore, a mutually reliant information ecology is established through the extensive proliferation of personal computing devices among customers, including gaming consoles, tablets, multifunctional workstations, cellphones, and others. An individual in the United States interacts with around ten technological gadgets per day on average [4]. This figure will increase to numerous dozens in the forthcoming decades. The emergence of the "Internet of Things" will catalyze a technological paradigm shift, characterized by the seamless integration of trillions of small devices and sensors into the information environment. This paradigm anticipates that 19.8 billion data-generating electronic gadgets will be connected to the internet, with exponential growth expected within this decade [5].

Artificial intelligence (AI) and machine learning (ML) are rapidly advancing fields, driving increased internet usage through AI-powered applications, cloud services, and the Internet of Things (IoT). All of these elements have played a part in the demand for ever-increasing speed, computing capacity, and power efficiency. Before 2007, human civilization could potentially compute 6.4×10^{18} digital commands per second and had saved 2.9×10^{20} digital bits. [6] The number of commands per second rose by 58% every year between 1986 and 2007. [4] Every year, the volume of data stored in huge electronic memories has grown by 23% [6]. The rates have increased considerably more since 2007. The total amount of data generated, recorded, and used worldwide in 2024 was 149 zettabytes. [7] It is anticipated to increase to 181 Zettabytes by the end of 2025. [7] For comparison, one zettabyte is equivalent to 10^{21} bytes. Essentially, it is anticipated that the rate of information generation and processing will rise at unthinkable speeds.

In addition to the requirement for constantly improving data computing skills, we also have to optimize the power consumption required for this reason. According to the International Energy Agency, data centers use between 2% and 3% of the world's electricity, or around 350 TW-h in 2023. [8] The DOE's Industrial Efficiency and Decarbonization Office provided

funding for the Center of Expertise for Energy Efficiency in Data Centers at Lawrence Berkeley National Laboratory to publish the ”2024 United States Data Center Energy Usage Report.” [9] The energy consumption of data centers from 2014 to 2028 is described in the paper. The paper states that the yearly energy usage of data centers in the United States stayed constant at about 60 TWh between 2014 and 2016. [9] This figure increased to approximately 76 TWh by 2018, representing 1.9% of the nation’s total electricity usage [9]. Data centers in the United States used 176 TWh of energy as of 2023, which accounted for 4% of the country’s total electricity consumption [9]. Data centers’ global power consumption could soon catch up to that of industrialized nations like Japan [8].

Digital ICs are used to process this confusingly massive amount of information. Since their creation in 1947, transistors have led the way in data processing. They are the fundamental components of all Integrated Circuits (ICs) that we can imagine, including digital logic circuits. They have continuously been reduced in size while also improving processing and power consumption. Gordon Moore proposed Moore’s law in 1965, which links an IC’s transistor count with its increase over time. According to this, an IC’s transistor count doubles every two years [10]. Moore estimates that between 1975 and the following ten years, the compound annual growth rate in this number of transistors would be 41% as shown in figure 1.1.

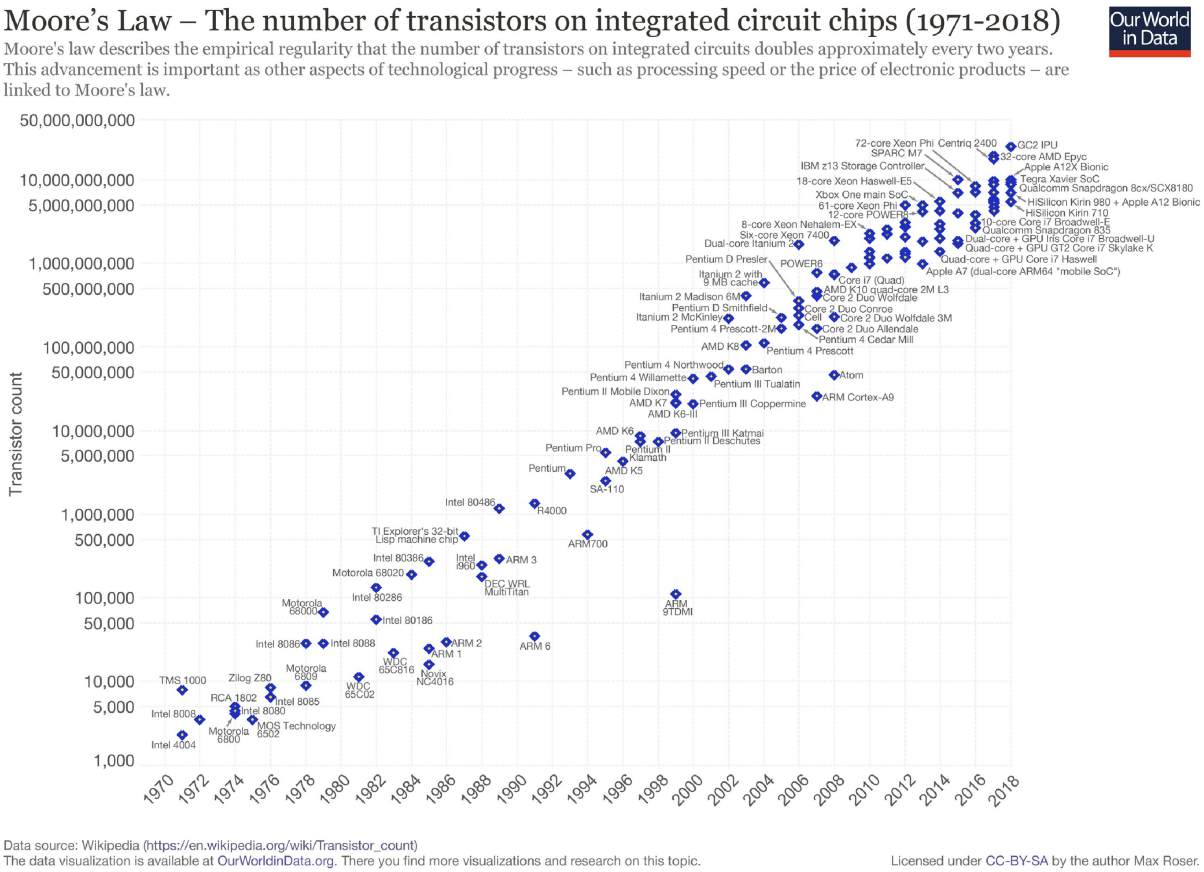


Figure 1.1: A semi-log plot of transistor counts for microprocessors against dates of introduction, according to Moore’s law [1]

Moore's second law, or Rock's law, argues that the expense of constructing a semiconductor chip fabrication facility doubles every four years. [11] In 2015, this price was 14 billion USD. [12]. This further substantiates Moore's first law, as the escalation in product cost correlates with the rise in transistors and miniaturization. Currently, the shrinking of transistors has reached the limits of existing silicon-based technology. As feature dimensions near atomic scales, such as 2 nm, numerous physical and economic constraints emerge. In tiny transistors, electrons may tunnel through the thin gate oxide, resulting in leakage current that diminishes battery lifespan and increases power usage. Moreover, as transistors decrease in size and increase in quantity, they must be densely packaged within the chip space, which negatively affects heat dissipation and thermal management. At the atomic scale, even minor fluctuations in the fabrication process might result in unstable transistor performance. Furthermore, in the atomic dimension domain, ballistic transport becomes more prominent, resulting in compatibility challenges with existing silicon technology.

The shrinking size of transistors presents challenges in heat management. "Dennard Scaling" posits that when transistor dimensions diminish, power consumption reduces approximately in proportion to the area. Consequently, the power density of the chip remains nearly invariant. Figure 1.2 illustrates that the power density level for CMOS half-pitch has remained approximately 100 W/cm². Elevating the working frequency will yield a power density beyond this threshold, which is impractical. Consequently, the operating frequency has been declining for several years, particularly following the advent of the 10 nm process, and we are unable to surpass the 3 GHz threshold.

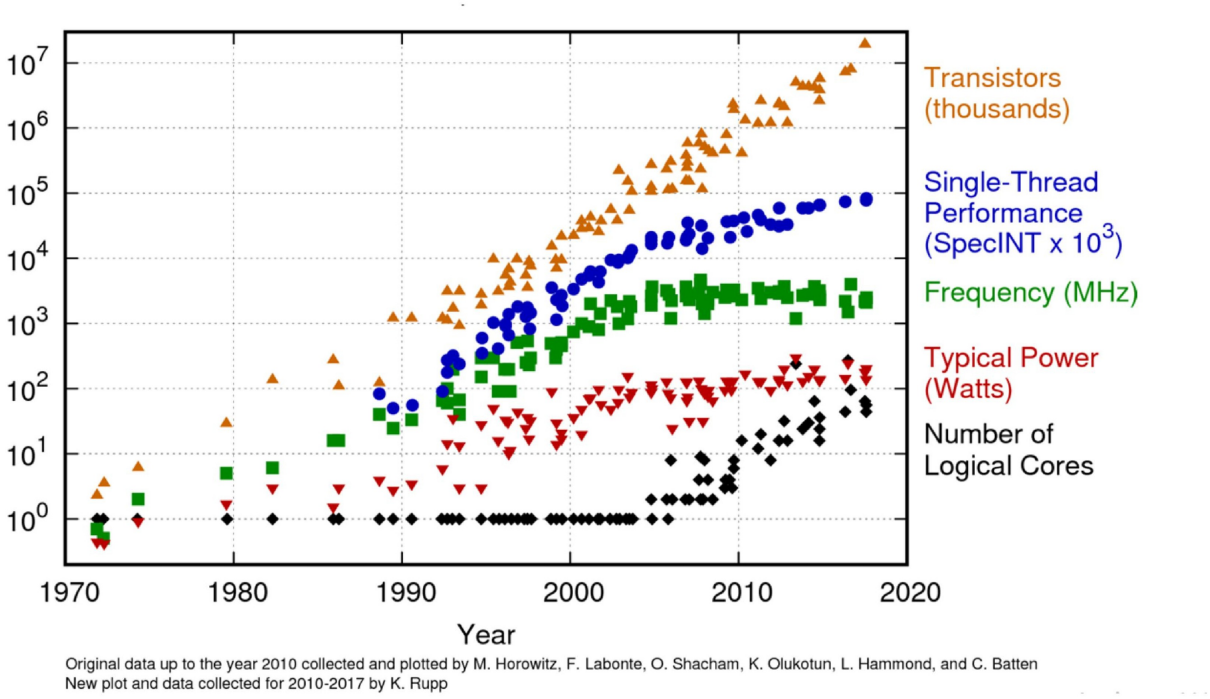


Figure 1.2: A summary of different metrics of microprocessors over 42 years

Two-dimensional, or single-layer, materials can be used to address these issues with the current silicon transistors. Two-dimensional materials are nanomaterials having a thickness of one or two atoms. Recently, the creation of graphene has demonstrated remarkable potential across different applications and attracted significant interest. They have emerged as promising materials because of their association with topological insulators. Certain 2D materials function as topological insulators, wherein the topological insulator, when restricted to two dimensions, displays conductive edge states while maintaining insulating bulk states. A fundamental attribute of 2D topological insulators is the Quantum Hall Effect. Robust spin-orbit coupling in these materials results in spin-polarized edge states, a phenomenon known as the Quantum Hall Effect, which facilitates electrical conduction without backscattering.

Graphene is a monolayer of carbon atoms [13] organized in a honeycomb form. It demonstrates elevated electron mobility at ambient temperature. [[14]- [15]] However, as its conduction band and valence band converge at the vertices of the hexagonal Brillouin zone, it exhibits no band gap. To open a bandgap in graphene, one must consider its nanoribbon configuration, which significantly reduces mobility and heightens vulnerability to edge roughness. [16] To overcome these drawbacks of using graphene as a channel of FET, 1T′-WTe_2 is utilized as a channel material. To the best of current knowledge, a TFET based on 2D 1T′-WTe_2 nanoribbons (NRs) has not yet been proposed. Therefore, a study focusing on designing an energy-efficient TFET based on room-temperature topological insulator 2D 1T′-WTe_2 NR is timely and necessary.

1.2 Objectives of the Thesis

The main objectives of this thesis are as follows:
1. To investigate the electronic properties (band structure, projected band structure, and surface states) of 1T′-WTe_2 with and without spin orbit coupling and to determine its topological property through calculation of topological invariant Z_2.
2. To analyze the electronic properties of 1T′-WTe_2 nanoribbons and to explore their transport characteristics, such as transmittance and current, considering pristine and defected TFET channels.
3. To investigate the electric field-induced topological phase transition in 2D 1T′-WTe_2 and assess critical TFET performance parameters, such as I-V characteristics, Ion/Ioff ratio, total capacitance, subthreshold swing, switching delay time, and dynamic power dissipation.
4. To evaluate the effect of using high-k dielectric material in gate on the performance of FET.

1.3 Thesis Outline

In the thesis, a topological insulator nano-ribbon is used as the channel material in a topological insulator FET model. Reducing power dissipation and promoting faster, more energy-efficient FET operation are the driving forces behind the effort. The five chapters that make up the dissertation are listed as follows:

Chapter 1: Introduction – This chapter describes the necessity of energy-efficient high-speed operation of FETs and an alternative way of achieving it using TIFETs.

Chapter 2: Literature Review – An outline of earlier studies on the characteristics and uses of topological materials is provided.

Chapter 3: Background – The fundamental physics of topological materials is dis cussed.

Chapter 4: Device Geometry and Simulation Methodology – The device geometry and simulation methodologies, incorporating density functional theory (DFT) and nonequilibrium Green's function (NEGF) formalism, are outlined.

Chapter 5: Results and Discussion –

A comprehensive examination of the electrical, topological, and transport characteristics of the $1T'$-WTe_2 monolayer and nanoribbons is presented. The transfer characteristics and performance of the proposed TIFET are also analyzed.

Chapter 6: Conclusion – The thesis's principal findings are described, and prospective areas for further research are discussed.

CHAPTER 2: LITERATURE REVIEW

2.1 Topological Insulator

Topological insulators (TI) are materials that function as standard insulators in their interior while supporting resilient, gapless states at their surfaces. These surface or edge states exhibit robustness to impurities and deformations, as they originate from the material's nontrivial topological order, which is frequently linked to robust spin-orbit coupling and time-reversal symmetry. Topological insulators possess prospective uses in spintronics, quantum computing, and innovative electronic devices owing to their capacity to sustain dissipationless edge or surface currents.

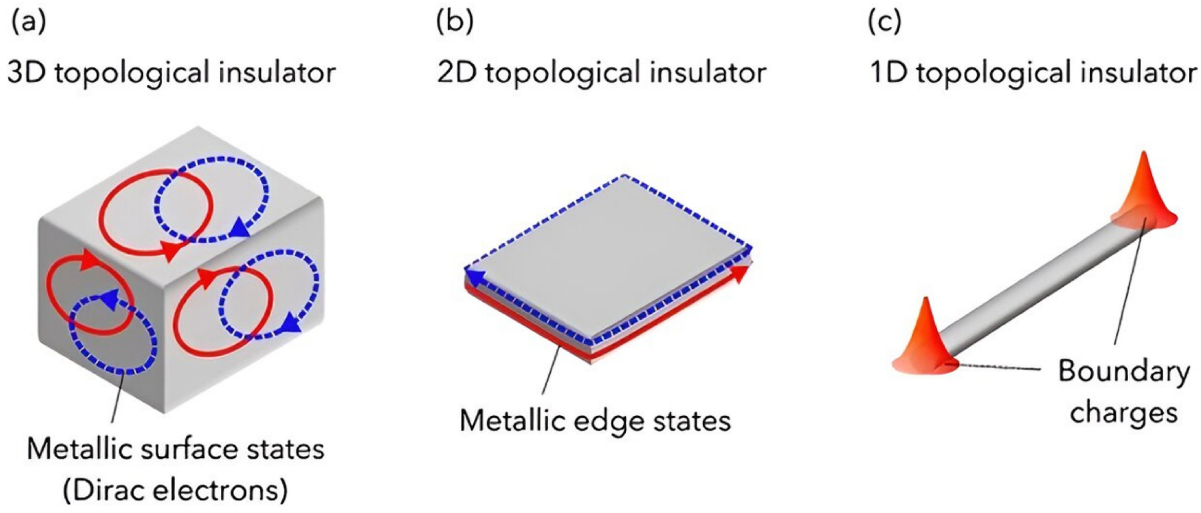


Figure 2.1: Schematics of (a) three-dimensional, (b) two-dimensional, and (c) one-dimensional topological insulators. This figure has been taken from Ref [2]

2.2 TI based FET

The interest in topological-insulator-based field-effect transistors (TIFETs) has increased, presenting a potential avenue for advancements in beyond-CMOS electronics. Topological transistors present multiple advantages compared to alternative transistor models that employ two-dimensional (2D) materials, including graphene and ballistic transistors. TI-FETs offer a significant advantage through their capacity to suppress backscattering, attributed to spin–momentum locking, leading to improved resistance to scattering and enhanced performance stability.

The topological phase transition is essential for an efficient switching mechanism, yielding a superior on-to-off current ratio relative to other nanoscale transistors. The core principle of a

topological transistor is the utilization of the disparity between the conducting state, facilitated by edge or surface channels, and the conventional insulating state. An electric field applied through a gate can, in principle, transition a material between a topological insulator phase and a trivial insulator phase, thereby regulating edge conduction [17].

This kind of device would work like a transistor, with a dissipationless topological channel in the ”on” state and an insulator in the ”off” state due to the bulk bandgap. Initial theoretical models indicated that a sufficiently strong electric field or applied strain could trigger a topological phase transition in specific materials, such as converting a 2D quantum spin Hall insulator into a conventional insulator. The TI-FET concept promises ultra-low power logic by enabling current flow through resistance-free edge paths in the “on” state, while completely suppressing it in the “off” state, thereby theoretically providing perfect on/off contrast.

Recent studies have advanced in the identification of appropriate materials for switching applications. Thin films of Na_3Bi , a three-dimensional Dirac semimetal, have been demonstrated to transition into two-dimensional topological insulators upon reduction to a monolayer, displaying a significant bandgap greater than 400 meV. Additionally, applying an electric field to this material can induce a transition to a trivial insulator state [18]. One of the most important achievements was found in Bi_2Se_3 nanowires, in which field-effect transistors exhibit extremely sharp turn-on behavior, low cutoff current, and high on/off current ratio indicating Keyes-like curves due to significant contribution by topological surface states whenapplying electric filed [19]. Another interesting experiment with $(Bi_xSb_{1-x})_2Te_3$ topological insulator nanoplates was ambipolar conduction including two carriers electron and hole which could be both varying by gate voltage, laying the foundation for that it is appropriate to make topological surface state gating in a FET structure [20]. This finding indicates that a practical topological transistor operating at room temperature may be achieved by choosing materials situated near the boundary between topological and trivial phases, and employing gating to switch the phase. Experimental developments like the electrical gating of WTe_2 or the electric field control of magnetism in magnetic TIs are quickly increasing the viability of topologically switchable devices, even though a fully functional TI-FET is still in the conceptual stage [21].

2.3 1T′-WTe_2 as Channel Material

Quantum Spin Hall (QSH) insulators, recognized as two-dimensional topological insulators (TIs), were first introduced in relation to graphene by C. L. Kane and E. J. Mele [22]. Subsequent investigations revealed that the band gap induced by spin-orbit coupling in graphene is quite minimal, posing challenges for the observation of the quantum spin Hall effect under existing experimental conditions [23]. The initial experimental validation of the QSH effect was achieved through HgTe/CdTe quantum wells, as demonstrated by Markus König, Steffen Wiedmann, and Christoph Brüne [24]. Nonetheless, these systems exhibit extremely narrow

bulk band gaps (on the order of meV) and necessitate ultra-high-quality samples along with ultra-low temperatures to detect the effect.

This study demonstrates that 1T′-WTe_2 serves as a viable foundation for the advancement of innovative topological electronic devices that may address the aforementioned challenges. Through first-principles calculations, we identify a novel category of large-gap $(> 0.1eV)$ quantum spin Hall insulators in the monolayer of 1T′-WTe_2. This study showcases the feasibility of an innovative topological field effect transistor utilizing 2D 1T′-WTe_2 nanoribbons. The device demonstrates an increased conductance through QSH edge channels when in the ON state and can be quickly turned OFF through a topological phase transition by the application of an electric field. The results of our electrically configurable QSH devices suggest promising applications in low-power nanoelectronics and could facilitate the development of practical architectures for topological quantum computing.

2.4 1T′-WTe_2 Structure

The monolayer of transition metal dichalcogenides (TMDC) exhibits a non-trivial QSH insulator ([25]- [26]). It has a two-dimensional crystalline structure. It comprises covalently bonded X-M-X sandwiches (M=Mo, W, Ta; X=Te, S, Se). Among these, 2H-WTe_2 possesses periodic hexagonal symmetry akin to that of graphite. Conversely, the distorted structure 1T′-WTe_2 has an orthorhombic configuration. 1T′-WTe_2 is more stable than 1T-WTe_2 [26]. Figure 2.2(a) illustrates the unit cell of the 1T′-WTe_2 two-dimensional (2D) monolayer from z-y axis. The blue represents W, the brown signifies Te, and the axes are designated in the x, y, and z directions. In a unit cell, there exist two W atoms and four Te atoms. The positions of the Te_1, Te_2, and W atoms are clearly marked, with the Te_1 atom positioned slightly below the Te_2 atom. Figure 2.2(b) presents the top view of the 1T′-WTe_2 monolayer, where the zigzag and armchair nanoribbons are indicated by red and black rectangles, respectively, which correspond to the arrangement of the W atoms along the edges of the monolayer [27]. The atomic structure is shown with atomic periodicity, highlighting the orientation and geometry of the two unique nanoribbon species. A side view of 1T′-WTe_2 monolayer is shown in figure 2.2(c), where the distorted structure of this material can easily be observed. The top and bottom Te atoms are not arranged in the same plane; they have a slight difference in atomic positions along the z-axis.

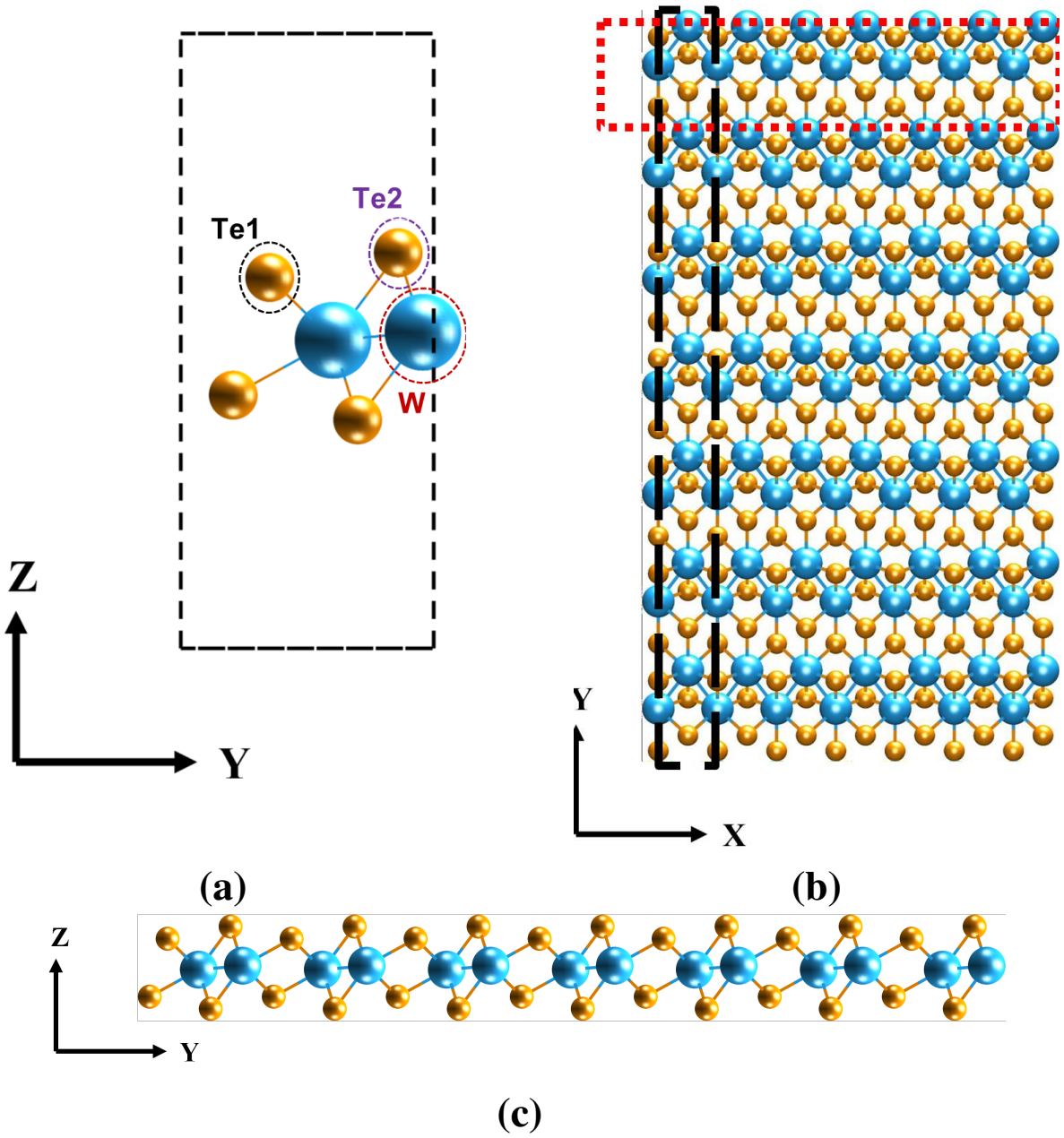


Figure 2.2: Schematic diagram of the atomic structure of (a) the unit cell of 1T′-WTe_2 bulk monolayer and (b) the top view of the monolayer 1T′-WTe_2.(c) side view of the monolayer 1T′-WTe_2.

CHAPTER 3: BACKGROUND

3.1 Historical Development of Topological Materials

In the 1980s, the quantum Hall effect (QHE) gave rise to the idea of topologically separate electronic phases. The integer Quantum Hall Effect (QHE), discovered in 1980, demonstrated that a two-dimensional electron gas subjected to a strong magnetic field can exhibit quantized Hall conductance, a phenomenon subsequently recognized as a topological invariant of the electronic band structure [28]. In 1982, Thouless et al. demonstrated that the quantized Hall conductance is associated with an integer Chern number, underscoring a profound topological origin [29]. These results established the foundation for "topological" phases of matter that extend beyond the traditional symmetry-breaking framework.

In the mid-1980s, theorists commenced speculating on Hall-like effects in the absence of magnetic fields. In a study published in 1985, B. A. Volkov and O. A. Pankratov posited that a band-inverted semiconductor junction may accommodate gapless two-dimensional electron states at its interface [30]. They demonstrated that a t the interface of a "inverted" semiconductor (such as HgTe or SnTe) and a conventional semiconductor, massless two-dimensional states emerge, anticipating the safeguarded surface states of topological insulators. This served as an initial indication of band structure, specifically attributed to inversion. Band order can generate conducting boundary states even without an external field. In 1988, few years later, F. D. M. Haldane established a significant milestone by developing a theoretical lattice model based on a graphene-like honeycomb lattice that realizes a quantized Hall effect conductivity in the absence of a net magnetic field. The Haldane model illustrated that the inherent violation of time-reversal symmetry, achieved through complex next-neighbor hopping, might yield a Chern insulator—characterized by an insulating bulk and chiral edge currents similar to those observed in the quantum Hall effect, even in the absence of an external magnetic field [31]. Haldane's research presented the concept of a topological band insulator influenced by lattice effects, demonstrating how a material may function as an insulator in t he predominant metallic edge states arise from a nontrivial band topology.

A significant advancement occurred in 2005 with the contributions of Charles L. Kane and Eugene J. Mele. Kane and Mele presented a novel topological quantum number to categorize time-reversal invariant band insulators in a pair of theoretical works [32]. They demonstrated that graphene, with substantial spin–orbit coupling, may theoretically sustain a quantum spin Hall effect, rendering it a two-dimensional topological insulator (2D TI). They specifically elucidated how spin–orbit coupling can induce a bandgap in graphene and produce helical edge states, while also identifying a Z2 topological order that differentiates this phase from a con-

ventional insulator. Despite graphene's minuscule spin–orbit gap, Kane and Mele's theoretical prediction was groundbreaking: it represented the inaugural explicit proposal of a quantum spin Hall insulator, a material exhibiting topologically protected edge states absent any magnetic field. This study established the Kane-Mele Z2 invariant as the mathematical basis for time-reversal-invariant topological insulators.

Simultaneously, a further theoretical advancement indicated a more experimentally viable pathway. In 2006, B. Andrei Bernevig, Taylor Hughes, and Shou-Cheng Zhang proposed that specific semiconductor quantum wells might demonstrate the Quantum Spin Hall Effect (QSHE) [24]. In their 2006 Science study, they projected that a mercury telluride/cadmium telluride (HgTe/CdTe) quantum well will experience a band inversion upon exceeding a certain thickness, therefore converting the system into a topological insulating state. This established the foundation for the initial experimental identification of a topological insulator. The theoretical predictions were remarkably validated in 2007. Markus Konig, Hartmut Buhmann, Laurens Molenkamp, and their colleagues at the University of Wurzburg constructed HgTe/CdTe quantum well devices and detected the quantized edge conductance indicative of the Quantum Spin Hall Effect [33]. This marked the inaugural experimental demonstration of a two-dimensional topological insulator, namely a quantum spin Hall insulator state. By modulating an external gate voltage, scientists were able to adjust the device from a conventional insulator (similar to CdTe, exhibiting no edge conductance) to a topological insulator (like to HgTe, characterized by edge conduction). The finding of strong, non-dissipative edge transport in the absence of a magnetic field was a significant milestone, demonstrating the existence of a novel phase of matter. The two-dimensional topological insulator in HgTe quantum wells is frequently referred to as the quantum spin Hall state, analogous to the quantum Hall state, but now safeguarded by time-reversal symmetry instead of a magnetic field.

With the availability of several 2D and 3D material systems, research into the special electrical characteristics of topological insulators and their potential applications exploded in the 2010s. A key characteristic is that topological insulator surface and edge states are intrinsically linked to electron spin. The one-dimensional edge channels in a two-dimensional topological insulator exhibit opposite spins traveling in opposite directions, creating a "helical" pair. Additionally, the two-dimensional surface of a three-dimensional topological insulator contains a Dirac fermion, with its spin orientation constrained to its direction of motion. An electron traveling a topological insulator surface exhibits a distinct spin orientation, and executing a 180° U-turn necessitates a spin flip, which is inhibited under conditions of unbroken time-reversal symmetry. The outcome demonstrates robust conduction on the surfaces of topological insulators; even in the presence of non-magnetic impurities or imperfections, electrons exhibit resistance to backscattering due to the absence of available states of the same spin for scattering. The robustness and intrinsic coupling between charge and spin currents indicate that topological insulators may serve as a platform for low-dissipation electronics and spintronics.

3.2 Spin-Orbit Coupling

The spin–orbit interaction is a relativistic effect resulting from the coupling of a particle's intrinsic spin with its motion in a potential ([34]- [35]). In atomic systems, electrons revolve around the nucleus. From the electron's reference frame, the positively charged nucleus seems to orbit around it. The relative motion of the nucleus produces a magnetic field as a result of the corresponding current, as anticipated by special relativity.

In general, an electron moving with velocity $\mathbf{v}$ in an electric field $\mathbf{E}$ perceives a magnetic field in its rest frame as a result of the Lorentz transformation:

$$\mathbf{B} = \frac{1}{c^2}\,\mathbf{v} \times \mathbf{E} \tag{3.1}$$

Substituting the velocity $\mathbf{v} = \frac{\mathbf{p}}{m}$, and assuming the electric field is radial $\mathbf{E} = E\,\hat{\mathbf{r}}$, we get:

$$\mathbf{B} = -\frac{\mathbf{r} \times \mathbf{p}}{mc^2}\,\frac{E}{r} \tag{3.2}$$

Now using $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ and expressing the electric field as $\mathbf{E} = -\nabla V$, where $V = \frac{U(r)}{e}$ is the electric potential corresponding to potential energy $U(r)$, we can rewrite:

$$\mathbf{B} = \frac{1}{mec^2} \cdot \frac{1}{r}\frac{dU(r)}{dr}\,\mathbf{L} \tag{3.3}$$

This shows that the relativistic magnetic field experienced by the electron is proportional to its orbital angular momentum L.

The spin magnetic moment of the electron is given by:

$$\boldsymbol{\mu}_s = -g_s \mu_B \frac{\mathbf{S}}{\hbar} \tag{3.4}$$

where $\mu_B = \dfrac{e\hbar}{2m}$ is the Bohr magneton, $\mathbf{S}$ is the spin angular momentum, and $g_s \approx 2.002319$ is the electron spin g-factor.

The interaction energy between a magnetic moment and a magnetic field is:

$$\Delta H = -\boldsymbol{\mu} \cdot \mathbf{B} \tag{3.5}$$

Substituting the relativistic magnetic field into this expression leads to the spin–orbit coupling Hamiltonian:

$$H_{\text{SOC}} = \frac{1}{2m^2c^2r}\frac{dV}{dr}\,\mathbf{L} \cdot \mathbf{S} \tag{3.6}$$

This term represents the additional potential energy resulting from spin–orbit coupling and plays a significant role in condensed matter systems.

3.3 Symmetry in Condensed Matter Systems

Symmetry is essential in defining the physical properties of condensed matter systems. Among the numerous symmetries, two of the most significant are time-reversal symmetry (TRS) and inversion symmetry ([36], [37]).

3.3.1 Inversion Symmetry

Inversion symmetry is a point symmetry that signifies the invariance of a system under point reflection. With respect to an inversion center, the crystal potential satisfies:

$$V(-\mathbf{r}) = V(\mathbf{r}). \tag{3.7}$$

The inversion operation is performed by the *parity operator* P. Its action on a wave function is:

$$P\psi(\mathbf{r}) = \psi(-\mathbf{r}). \tag{3.8}$$

If $P\psi(\mathbf{r}) = \psi(\mathbf{r})$, the state has *even parity* with eigenvalue $+1$; if $P\psi(\mathbf{r}) = -\psi(\mathbf{r})$, it has *odd parity* with eigenvalue -1. In crystals possessing inversion symmetry and spin–orbit coupling (SOC):

$$\psi_{n,\mathbf{k},\chi}(-\mathbf{r}) = \psi_{n,-\mathbf{k},\chi}(\mathbf{r}). \tag{3.9}$$

The electronic energy bands obey:

$$E_{n,\chi}(-\mathbf{k}) = E_{n,\chi}(\mathbf{k}). \tag{3.10}$$

This implies that the energy of an electron at momentum k for spin up/down equals that at $-k$.

A simple example is a particle in an infinite potential well (Fig. 3.1). Here, with respect to the well's center:

$$V\Big(-\frac{a}{2}\Big) = V\Big(\frac{a}{2}\Big) = \infty. \tag{3.11}$$

The stationary states,

$$\psi(x) = \sqrt{\frac{2}{a}}\sin\Big(\frac{n\pi x}{a}\Big), \tag{3.12}$$

exhibit even and odd parity due to inversion symmetry.

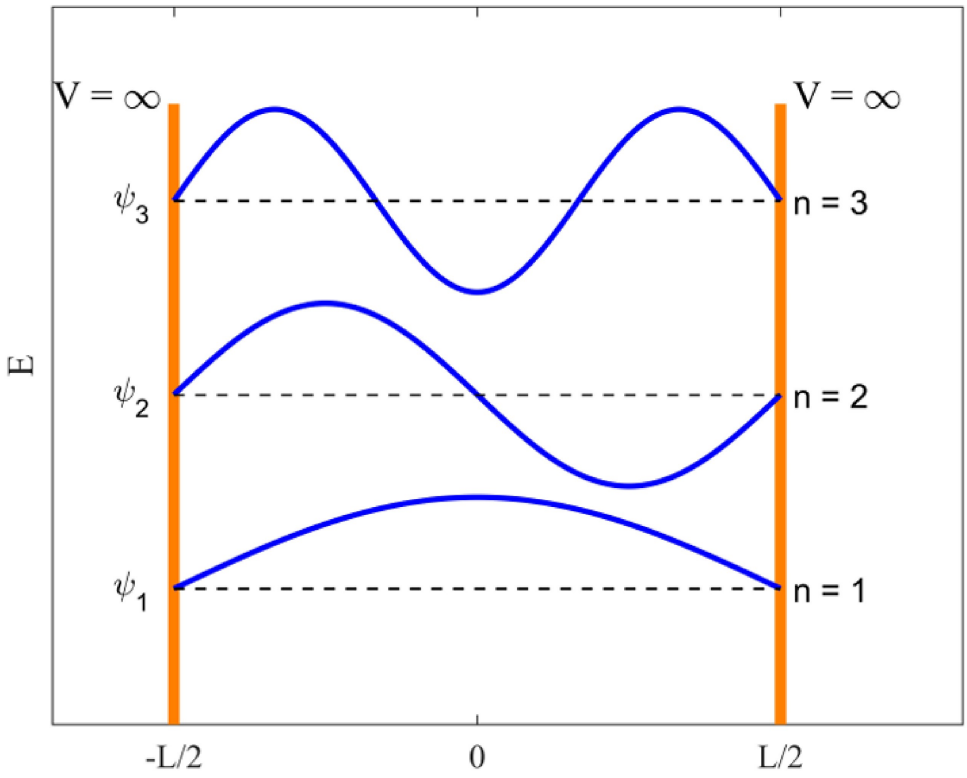


Figure 3.1: Stationary states of electrons in an infinite potential well. The wave functions exhibit even and odd parity as a consequence of Inversion symmetry.

3.3.2 Time-Reversal Symmetry

Time-reversal symmetry (TRS) refers to the invariance of physical laws under the transformation $t \to -t$. In the absence of magnetic fields or impurities, a system can preserve TRS; however, TRS is broken in the presence of a magnetic field.

In quantum systems, if TRS holds:

$$\mathcal{T}H\mathcal{T}^{-1} = H. \tag{3.13}$$

For spin-$\frac{1}{2}$ particles, the TRS operator flips spin and complex conjugates the wave function. The energy bands satisfy:

$$E_{n,-\chi}(-\mathbf{k}) = E_{n,\chi}(\mathbf{k}), \tag{3.14}$$

meaning the energy of an up/down spin electron at $\mathbf{k}$ equals that of a down/up spin electron at $-\mathbf{k}$. If both TRS and inversion symmetry are present, bands are doubly degenerate:

$$E_{n,-\chi} = E_{n,\chi}. \tag{3.15}$$

3.4 Topological Invariant

Topology is a mathematical concept. In mathematics, topology is the study of the properties of shapes, spaces and objects that remain unchanged under continuous transformations like stretching, bending or twisting—but not tearing or gluing. A topological space is a set of points with a structure that tells which points are close to each other but without distances, unlike geometry [38]. A doughnut and a coffee mug are topologically the same because one can be stretched or bent into the other without cutting or gluing (Fig. 3.2). These kinds of objects are called homeomorphic and this property is called homeomorphism, which is defined as a continuous function $f : X \to Y$ with a continuous inverse f^{-1}. Topological invariants are properties that remain unchanged under homeomorphism. One particular topological invariant is the genus, which represents the number of holes in an object. For instance, a sphere has genus 0 (no hole) and a torus has genus 1 (one hole).



(a) A doughnut (b) A coffee mug

Figure 3.2: Illustration of topological equivalence: A doughnut (left) and a coffee mug (right) are topologically equivalent objects, as both have a genus of 1 (i.e., a single hole). They can be continuously deformed into one another without cutting or gluing, demonstrating the concept of homeomorphism in topology.

How can we differentiate materials based on topology? The answer lies in the band theory of solids. In a periodic crystal, the potential energy experienced by an electron is periodic due to the periodic arrangement of atoms $V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$, where $\mathbf{R}$ is the Bravais lattice vector. According to Bloch's theorem the electron wavefunction in a periodic crystal can be written as

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\, u_{\mathbf{k}}(\mathbf{r}) \tag{3.16}$$

where $u_{\mathbf{k}}(\mathbf{r})$ is periodic with the same periodicity as the lattice and $\mathbf{k}$ is the crystal momentum.

The Bloch Hamiltonian $H(\mathbf{k})$ is obtained via the Fourier transform of the real-space Hamiltonian. A band structure evolves by mapping the crystal momentum $\mathbf{k}$ to the energy eigenvalues. The corresponding eigenstates are the Bloch states $u(\mathbf{k})$. The topology of this Bloch bundle gives rise to topological invariants.

$$H(\mathbf{k})\, u(\mathbf{k}) = E(\mathbf{k})\, u(\mathbf{k}) \tag{3.17}$$

According to the Copenhagen interpretation of quantum mechanics, the wavefunction ψ itself does not have any direct physical significance; instead, $|\psi|^2$ represents the probability density of finding a particle, such as an electron. All measurable physical quantities involve $|\psi|^2$. Therefore, one might consider the phase of the wavefunction to be arbitrary and physically irrelevant. However, in 1984, Michael Berry introduced the concept of a geometric phase, now known as the Berry phase, which carries important physical implications [39].

Suppose a quantum system is described by a Hamiltonian $H(t)$ that depends smoothly on a set of external parameters $R(t)$. For instance, consider an infinite potential well whose width is adiabatically (slowly) varied from a to $2a$ and back to a. In such a scenario, the wavefunction remains in the instantaneous eigenstate but acquires an additional phase factor. Here, the well width acts as the parameter space.

By the adiabatic theorem, if the Hamiltonian $H(R(t))$ changes slowly, the system remains in the instantaneous eigenstate $\psi_n(R(t))$ up to a phase factor:

$$|\psi_n(t)\rangle = e^{i\theta_n(t)}\, |\psi_n(R(t))\rangle. \tag{3.18}$$

From the time-dependent Schrödinger equation

$$H(R(t))\, |\psi_n(t)\rangle = i\hbar \frac{d}{dt} |\psi_n(t)\rangle, \tag{3.19}$$

We substitute the above ansatz:

$$H(R(t))\, |\psi_n(R(t))\rangle = i\hbar \left[i\, \frac{d\theta_n(t)}{dt}\, e^{i\theta_n(t)} |\psi_n(R(t))\rangle + e^{i\theta_n(t)} \frac{d}{dt} |\psi_n(R(t))\rangle \right]. \tag{3.20}$$

Taking the inner product with $\langle \psi_n(R(t))|$ and using the normalization condition $\langle \psi_n(R(t))|\psi_n(R(t))\rangle = 1$, we get

$$E_n = -\hbar \frac{d\theta_n(t)}{dt} + i\hbar \langle \psi_n(R(t))| \frac{d}{dt} \psi_n(R(t))\rangle, \tag{3.21}$$

which leads to the differential equation for the phase:

$$\frac{d\theta_n(t)}{dt} = -\frac{E_n}{\hbar} + i \langle \psi_n(R(t))| \frac{d}{dt} \psi_n(R(t))\rangle. \tag{3.22}$$

Integrating over time, the total phase θ_n can be expressed as

$$\theta_n = -\frac{1}{\hbar}\int_0^t E_n(t')\,dt' + i\int_0^t \langle\psi_n(t')|\frac{d}{dt'}\psi_n(t')\rangle\,dt'. \tag{3.23}$$

The first term is known as the dynamic phase, and the second term is the Berry phase or geometric phase.

The first term is known as the dynamic phase, and the second term is the Berry phase or geometric phase.

Using the chain rule to express the time derivative in terms of the parameter R, we obtain the Berry phase as a line integral in parameter space:

$$\theta_{\text{geom}} = i\int_{R_i}^{R_f} \langle\psi_n(R)|\nabla_R\psi_n(R)\rangle\cdot dR. \tag{3.24}$$

The quantity

$$\mathcal{A}_n(R) = \langle\psi_n(R)|\nabla_R\psi_n(R)\rangle \tag{3.25}$$

is called the *Berry connection*. For crystalline solids, the parameter space is the crystal momentum $\mathbf{k}$ in the Brillouin zone, and the Berry connection is defined in momentum space as [40], [41]:

$$\mathcal{A}_n(\mathbf{k}) = \langle u_n(\mathbf{k})|\nabla_{\mathbf{k}}u_n(\mathbf{k})\rangle, \tag{3.26}$$

where $u_n(\mathbf{k})$ is the periodic part of the Bloch function. where $u_n(\mathbf{k})$ is the periodic part of the Bloch function.

The Berry connection is analogous to the electromagnetic vector potential $\mathbf{A}$, but it exists in momentum space. The Berry curvature, analogous to the magnetic field, is defined as the curl of the Berry connection:

$$\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}}\times\mathcal{A}_n(\mathbf{k}). \tag{3.27}$$

The Berry phase accumulated over a closed loop C in $\mathbf{k}$-space is given by the integral

$$\gamma_n = \oint_C \mathcal{A}_n(\mathbf{k})\cdot d\mathbf{k}. \tag{3.28}$$

Using Stokes' theorem, this can be related to the integral of the Berry curvature over the enclosed area S:

$$\gamma_n = \int_S \Omega_n(\mathbf{k})\cdot d^2k. \tag{3.29}$$

The Chern number c_n is a topological invariant defined as

$$c_n = \frac{1}{2\pi} \int_{\mathrm{BZ}} \Omega_n(\mathbf{k}) \cdot d^2k, \tag{3.30}$$

where the integral is over the entire Brillouin zone (BZ). A nonzero Chern number indicates that the system is a Chern insulator, which hosts robust chiral edge states protected against smooth perturbations as long as the bulk energy gap remains open. The Chern number counts how many times the eigenstates of the system twist as one traverses the closed manifold of the BZ.

where the integral is over the entire Brillouin zone (BZ). A nonzero Chern number indicates that the system is a Chern insulator, which hosts robust chiral edge states protected against smooth perturbations as long as the bulk energy gap remains open. The Chern number counts how many times the eigenstates of the system twist as one traverses the closed manifold of the BZ.

In the integer quantum Hall effect, each Landau level contributes a Chern number $c = 1$, resulting in the quantized Hall conductivity:

$$\sigma_{xy} = \frac{e^2}{h}\, c, \tag{3.36}$$

where c is the total Chern number. Materials with strong SOC and broken TRS, for example, due to magnetization, can exhibit the quantum anomalous Hall (QAH) effect. A QAH insulator is a two-dimensional (2D) insulating system that exhibits a quantized Hall conductivity without an external magnetic field. The edge states in a QAH insulator are chiral, with electrons propagating in only one direction along the sample boundary. The Hall voltage arises from the directional asymmetry of edge transport even in the absence of an external magnetic field.

If the system has Chern number zero, one must compute the $\mathbb{Z}_2$ invariant to characterize its topology. The $\mathbb{Z}_2$ index distinguishes between a trivial insulator and a quantum spin Hall insulator (QSHI) [42].

In quantum systems with TRS and half-integer spin, every energy eigenstate is at least two-fold degenerate, forming Kramers pairs. For systems with inversion symmetry, the $\mathbb{Z}_2$ invariant can be computed using the Fu–Kane formula, which uses the parity eigenvalues $\xi_{2m}(\Lambda_i) = \pm 1$ at time-reversal invariant momenta (TRIM) points Λ_i in the Brillouin zone [43]. In two dimensions, these high symmetry points are $\Gamma(0,0)$, $X(\pi,0)$, $Y(0,\pi)$, and $M(\pi,\pi)$.

The $\mathbb{Z}_2$ invariant ν is given by

$$(-1)^{\nu} = \prod_{i=1}^{4} \delta_i, \tag{3.37}$$

where

$$\delta_i = \prod_{m=1}^{N_{\mathrm{occ}}} \xi_{2m}(\Lambda_i), \tag{3.38}$$

and N_{occ} is the number of occupied Kramers pairs.

For systems lacking inversion symmetry, the Wannier charge center evolution method is used to extract the $\mathbb{Z}_2$ index [44].

A QSHI has an insulating bulk but supports helical edge states protected by TRS, in which electrons with opposite spin propagate in opposite directions. Unlike QAH insulators, the Hall conductivity of a QSHI is zero and no Hall voltage appears in the absence of a magnetic field. The $\mathbb{Z}_2$ index counts whether the system has an odd number of Kramers pairs of edge states.

CHAPTER 4: SIMULATION METHODOLOGY AND DEVICE GEOMETRY

4.1 First principal calculation

Our investigation thoroughly examined the geometric relaxation and electronic properties of the studied system using density-functional theory [45] calculations conducted using Quantum Espresso (QE) [46]- [47], employing the generalized-gradient approximation [48] parameterised by hybrid functional (HSE06) [49] with Optimized Norm-Conserving Vanderbilt (ONCV)pseudopotential [50]. The Kohn-Sham single-particle wavefunctions were expressed using a set of plane wave bases, with a kinetic energy cutoff of 680 eV. A 10×10×1 Monkhorst-Pack k-point grid was used for the Brillouin zone sampling. We employed a vacuum region exceeding 15 Å to guarantee adequate separation between the periodic pictures of 1T′-WTe_2. We initially analyse the structures of the monolayer optimised using HSE06, and including spin-orbit coupling (SOC). The optimised lattice parameters a, b, and c are 3.497 Å, 6.338 Å, and 15.431 Å, respectively.

4.2 Wannierization

Our aim is to formulate a tight-binding-like Hamiltonian for the purpose of developing a device Hamiltonian to evaluate transport features, necessitating the acquisition of localized, atomic-like orbitals in real space. Nonetheless, quantum espresso inherently yields extended Bloch wavefunctions in real space, which are confined in reciprocal (k-) space.

According to Bloch's theorem, the electronic eigenstates in a periodic crystal can be expressed as:

$$\psi_{nk}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\, u_{nk}(\mathbf{r}) \tag{4.1}$$

where:
– $\psi_{nk}(\mathbf{r})$ is the Bloch function for band n and wavevector $\mathbf{k}$,
– $e^{i\mathbf{k}\cdot\mathbf{r}}$ is a plane-wave factor, and
– $u_{nk}(\mathbf{r})$ is a lattice-periodic function satisfying $u_{nk}(\mathbf{r}+\mathbf{R}) = u_{nk}(\mathbf{r})$.

To transform these k-localized Bloch states into real-space localized functions, we used the WANNIER90 package [51–53], which performs a discrete Fourier transformation of the Bloch functions into WFs:

$$w_n(\mathbf{r}-\mathbf{R}) = \frac{1}{N_k}\sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}}\psi_{n\mathbf{k}}(\mathbf{r}) \tag{4.2}$$

Here, $w_n(\mathbf{r} - \mathbf{R})$ is the WF centered at lattice vector $\mathbf{R}$, and N_k is the total number of k-points sampled in the Brillouin zone. These WFs are localized in real space and can be regarded as atomic-like orbitals, making them suitable for tight-binding model constructions.

Our goal is to develop an accurate tight-binding model utilizing MLWFs that can precisely recreate the electronic band structure within the relevant energy range, particularly the bands at the Fermi level. High-quality WFs must meet the following four criteria:
1. Precise recreation of the DFT band structure within the designated energy range.
2. Maintenance of atomic orbital symmetry.
3. Excellent spatial localization.
4. Employ the minimal quantity of WFs required for the specified energy range.

To meet these criteria, three essential steps must be followed:
1. Select appropriate projectors.
2. Define a disentanglement energy window.
3. Define a frozen energy window.

We began by selecting projectors based on the knowledge of the orbital characteristics of the DFT bands of our system, 2D 1T′-WTe_2. The knowledge of these orbital characteristics is crucial for selecting suitable projectors and setting proper disentanglement and frozen energy windows. As shown in Figure 4.1, the projected band structures of 1T′-WTe_2 reveals that bands around the Fermi level (from -8 to 3.75 eV) are predominantly contributed by d orbitals for W atoms and p orbitals for Te atoms. Based on this, we selected a disentanglement energy window from -8 to 3.75 eV and used d orbitals for W atoms and p orbitals for Te atoms.

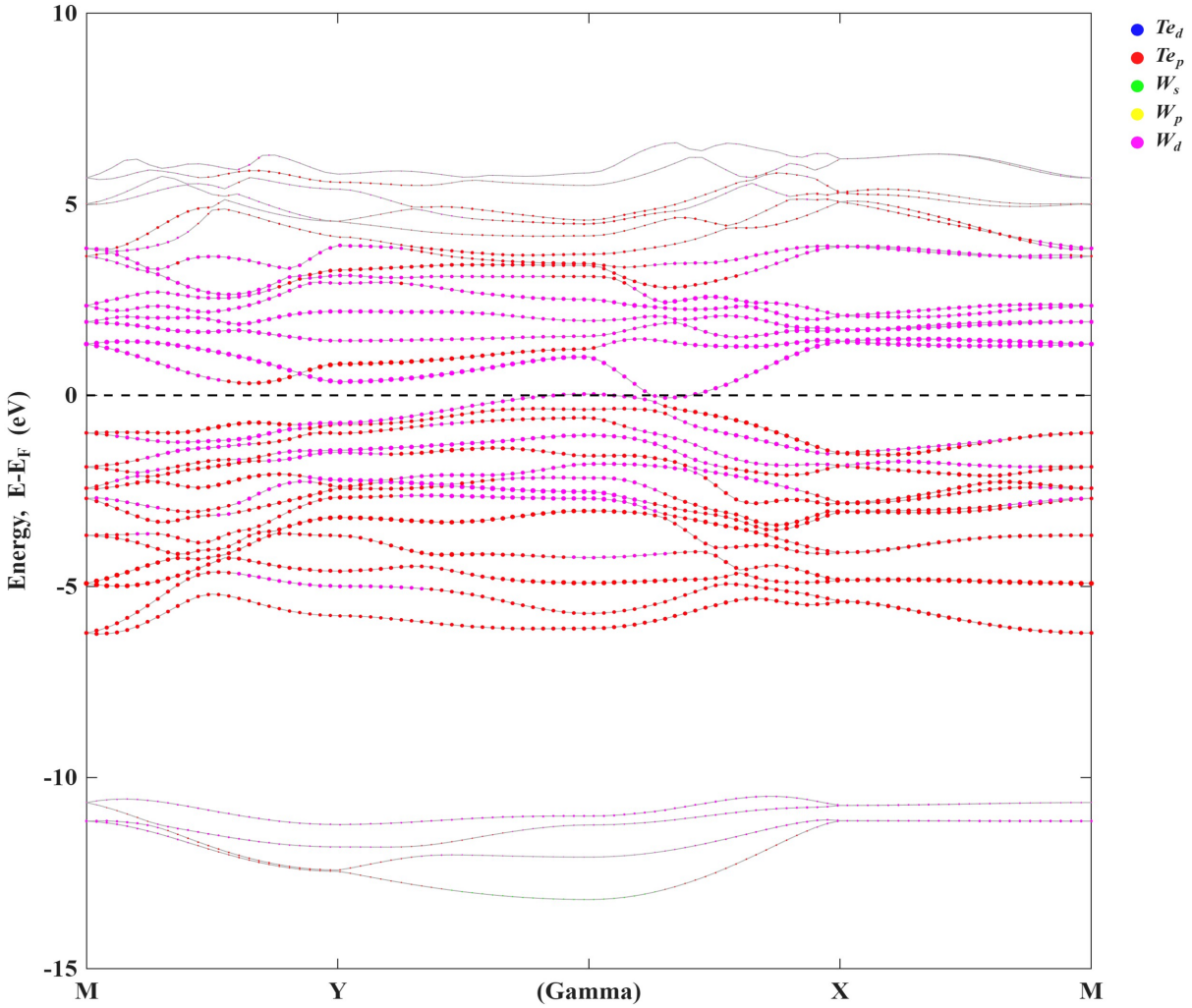


Figure 4.1: Electronic band structure of 2D 1T′-WTe_2, showing the contributions of d orbitals for W atoms and p orbitals for Te atoms to the bands near the Fermi level.

It is important to note that the number of bands within the disentanglement window must be at least equal to the total number of projectors multiplied by the number of atoms in the system. In our case:

1. 10 projectors per atom for W and 6 projectors per atom for Te (since with SOC, each orbital splits into two components: spin-up and spin-down)
2. Four Te atoms and two W atoms. Six atoms in a unit cell.

This exactly matches the number of bands within our selected energy window. To achieve atomic-like WFs, we set `num_iter` ≤ 20 in Wannier90. A larger number of iterations may cause excessive mixing between WFs, making them deviate from atomic-like shapes. Finally, it is worth recalling that the inverse Fourier transform allows the reconstruction of Bloch functions from WFs, using:

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}}\, w_n(\mathbf{r}-\mathbf{R}) \tag{4.3}$$

Wannier90 [53] is used to convert extended Bloch states to Maximally localized Wannier func-

tions (MLWFs). The MLWFs are employed to construct the tight-binding (TB) Hamiltonian which has also been utilized to study the topological properties of the 1T′-WTe_2 by calculating the topological invariant, $\mathbb{Z}_2$. The $\mathbb{Z}_2$ invariant effectively reflects the curvature information of the band structure in k -space, which is an expression of the topology of electronic materials. The calculation consists in finding the parity of the Wannier charge centers (WCCs) crossing at the middle of the Brillouin zone. The hybrid WCCs are written as [54]:

$$\bar{X}(k_2) = \frac{a_1}{2\pi} \int dk_1 A_n(k) \tag{4.1}$$

$A_n(k)$ is a berry potential and it can be written as [53]:

$$A_n(k) = i < u_{nk} | \nabla_k | u_{nk} > \tag{4.2}$$

Berry curvature is the gauge field arising from the berry potential. It is defined as [54]:

$$\mathbf{F} = \nabla_k \times A_n(k) \tag{4.3}$$

Topological invariant, $\mathbb{Z}_2$ can be calculated from the following equation [55]:

$$\mathbb{Z}_2 = \sum_{n}^{\text{OCC}} \frac{1}{2\pi} \left(\oint_{\text{Half BZ}} \mathbf{A}_n \cdot d\mathbf{k} - \int_{\text{Half BZ}} dk_x dk_y F_n \right) \tag{4.4}$$

$$= 0 \text{ or } 1 \pmod 2 \tag{4.5}$$

In this context, OCC refers to the occupied states, while k denotes the reciprocal lattice vector.

4.3 Generation of Tight-Binding-like Hamiltonian

To construct a tight-binding-like Hamiltonian, two essential files are required: `hr.dat` and `centres.xyz`. The `hr.dat` file contains the raw matrix elements of the WFs, while `centres.xyz` lists the WF centers in Cartesian coordinates.

In our system, each WF can spread over N nearest-neighbor images of the home unit cell, which is located at the lattice vector $\mathbf{R} = (0, 0, 0)$. For instance, Figure 4.2 illustrates the spatial spread of WFs in 2D WTe_2, extending over four neighboring images of the home unit cell. The first step in building the Hamiltonian involves generating bond interaction matrices between atoms. If atom 1 is associated with m WFs, and atom 2 is associated with n WFs, then the interaction between them is represented by an $m \times n$ matrix.

For instance, in the 2D WTe_2 system:
– With 6 atoms per unit cell.
– Assuming `spinor = .false.` (or `.true.`), resulting in 3×3 (or 6×6) matrices for

each bond interaction.

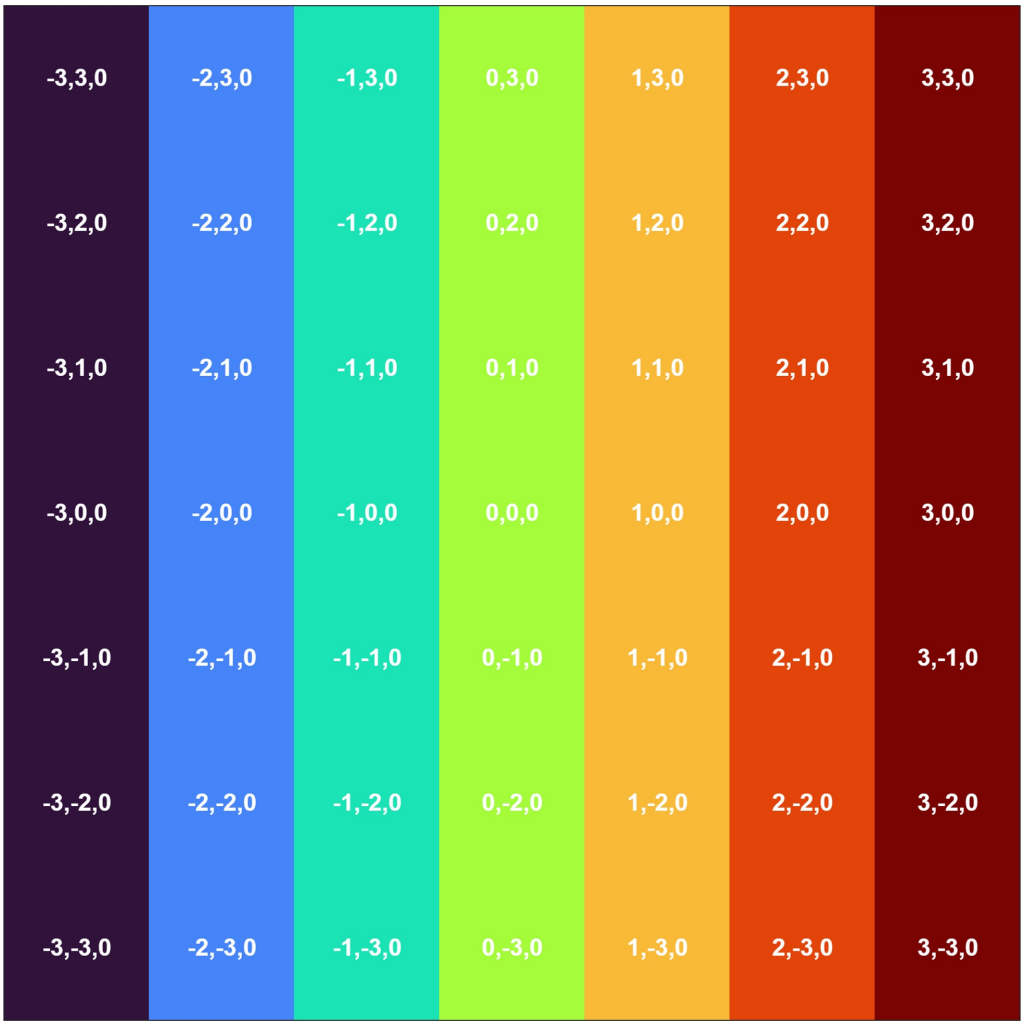


Figure 4.2: Depiction of WF spreads for 2D WTe_2.

There will be 256 such interaction matrices for each translational vector, $\mathbf{T}$.

If r is the index number of $\mathbf{T}$, and i, j vary from 1 to N, then the interaction matrix can be extracted from $\mathbf{H}_S$ as $h_{r,m}$, where $m = i \times j$.

To obtain the real-space tight-binding Hamiltonian matrix $\mathbf{H_R}$ positioned at lattice vector $\mathbf{R}$, we use the following algorithm:

1. Two nested loops iterate through i (from 1 to N) and j (from 1 to N).

2. r' is the index number of $\mathbf{R}$, which determines the first index of the matrix $\mathbf{H}_S$.

3. The index $m = i \times j$ determines the second index of $\mathbf{H}_S$, giving the required interaction matrix $h_{r',m}$.

4. At the block position (i, j) of $\mathbf{H_R}$, the interaction matrix $h_{r',m}$ is placed.

The band structure of bulk 2D-WTe_2 can be calculated using the k-dependent Hamiltonian $\mathbf{H}(\mathbf{k})$, which follows:

$$\mathbf{H}(\mathbf{k}) = \sum_{\mathbf{R}} \mathbf{H}_{\mathbf{R}}\, e^{i\mathbf{k}\cdot\mathbf{R}} \tag{4.4}$$

The band eigenvalues and eigenvectors satisfy:

$$\mathbf{H}(\mathbf{k})\,|\psi_{n\mathbf{k}}\rangle = E_{n\mathbf{k}}\,|\psi_{n\mathbf{k}}\rangle \tag{4.5}$$

where $E_{n\mathbf{k}}$ and $|\psi_{n\mathbf{k}}\rangle$ are the eigenvalue and eigenvector for band index n and wavevector $\mathbf{k}$, respectively.

4.4 Device geometry

We considered a double-gate structure for the 1T′-WTe_2 nanoribbon TFET, as shown in figure.4.3. Hafnium dioxide HfO_2 is used as the insulating layer between the gate and the channel region. The thicknesses of the nanoribbon and HfO_2 are $t_{\mathrm{WTe_2}} = 4.1938$ Å and $t_{\mathrm{HfO_2}} = 14$ nm [56], respectively. The dielectric constant for HfO_2 is $\epsilon_{HfO_2} = 25$ [57], while the dielectric constant for the 1T′-WTe_2 nanoribbon along the vacuum direction is assumed as $\epsilon_{\mathrm{WTe2}} = 1$.

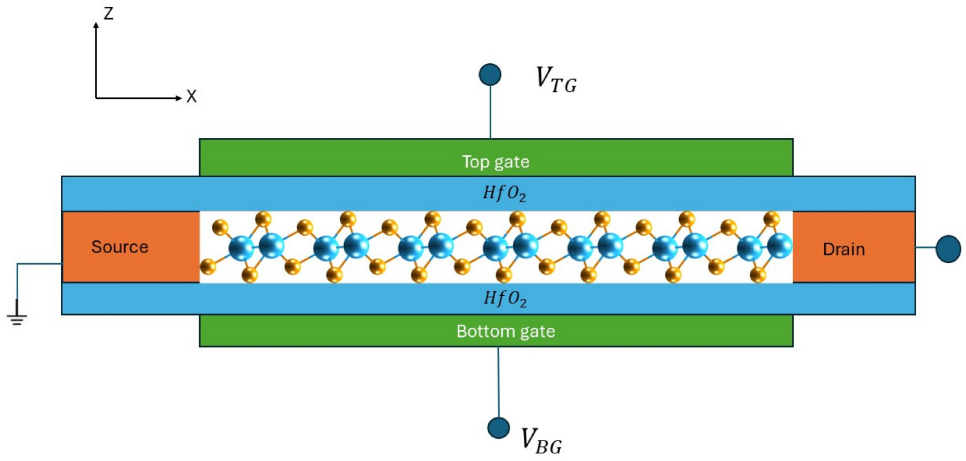


Figure 4.3: Side view of the double gate device geometry.

The suggested device architecture, which includes a monolayer 1T′-WTe_2 nanoribbon that is completely surrounded by top and bottom HfO_2 gate dielectrics, can be used with current two-dimensional (2D) materials processing technologies. Researchers have already made monolayer 1T′-WTe_2 in the lab using molecular beam epitaxy (MBE) on bilayer graphene/SiC and other van der Waals substrates. The result was high-quality epitaxial films with the right distorted octahedral structure and topological phase formation. Tang et al. showed that they could

grow stable monolayer 1T′-WTe_2 on bilayer graphene/SiC using MBE. The interfaces were atomically sharp and the electrical performance was strong [58]. Fei et al. and Naylor et al. have both reported similar MBE growth of monolayer WTe_2, which shows that wafer-scale epitaxial WTe_2 monolayers can be made over and over again [59], [60]. These methods of growth make it possible to directly move monolayer WTe_2 flakes using either PMMA-assisted wet transfer or van der Waals dry-pickup methods. The bottom-gate electrode can be patterned on a Si/SiO_2 substrate using standard lithographic techniques, followed by conformal deposition of a high-k HfO_2 dielectric layer via atomic layer deposition (ALD). People use ALD HfO_2 a lot to make 2D FETs because it is very uniform, has a low defect density, and works well with low-temperature processes. McDonnell et al. and Kim et al. showed that ALD HfO_2 can be evenly deposited on TMD surfaces at temperatures below 200 °C. This is especially true when thin metal seed layers or ozone-based surface functionalization are used to encourage nucleation [61], [62]. These steps make it possible to make a high-quality bottom HfO_2 dielectric that works perfectly with the next step of transferring WTe_2. While the monolayer WTe_2 is moved to the bottom HfO_2, the channel can be made into a nanoribbon using electron-beam lithography and then low-damage plasma or ion-beam etching. This is similar to how $MoTe_2$, MoS_2, and bismuthene nanoribbons are made.

1T′-WTe_2 is sensitive to air, so it needs to be encapsulated right away. After either mild ozone functionalization or the deposition of an ultrathin (0.5 nm) HfO_2 seed layer to ensure uniform nucleation, ALD can be used to deposit the top HfO_2 dielectric. Many people have used this method to protect 2D materials that are sensitive to air, like $MoTe_2$ and black phosphorus [63]. Finally, the top gate electrode can be shaped and put directly on top of the top HfO_2 layer, which finishes the dual-gate stack.

4.5 Formation of Full Device Hamiltonian

The device Hamiltonian $\mathbf{H}_{\text{Div}}$ can be generated by first creating a Hamiltonian for a building block of certain length and width, then repeating it along the transport direction. This approach allows for constructing a Hamiltonian matrix for devices of any length in a block tridiagonal manner.

If a building block consists of X unit cells in width and Y unit cells in length, the building block Hamiltonian and full device Hamiltonian can be created using the following algorithms:

Algorithm for Creating Building Block Hamiltonian $\mathbf{H}_{\text{BB}}$

1. Create a list of length XY to store the translational vectors $\mathbf{R}$ for the building block, and another list of the same length to store nearest-neighbor vectors $\mathbf{R}'$.

2. Use three nested loops over $r = 1 \ldots XY$, $i = 1 \ldots N$, and $j = 1 \ldots N$.

3. Compute $r_s = |r' - r|$, where r' is the index of $\mathbf{R}'$. Define a parameter W_{NN}, which sets the maximum nearest-neighbor range (e.g., $W_{\mathrm{NN}} = 2$).

4. The index $m = i \times j$ locates the required matrix $h_{r,m}$ in $\mathbf{H}_S$.

5. Place $h_{r,m}$ in the block position (i, j) of $\mathbf{H}_{\mathrm{BB}}$.

Algorithm for Coupling Matrix U_{BB}

To obtain the coupling matrix $\mathbf{U}_{\mathrm{BB}}$ between consecutive building blocks:

- Use the same nested loops.
- The translational vector is now $\mathbf{R}$ shifted by Y unit cells.
- Do not place matrices at positions where $i = j$.

Algorithm for Full Device Hamiltonian $\mathrm{H}_{\mathrm{Div}}$

1. Set the diagonal blocks: $A_{i,i} = \mathbf{H}_{\mathrm{BB}}$ for $i = 1, 2, \ldots, N$.

2. Set the off-diagonal couplings: $A_{i,i+1} = A_{i+1,i} = \mathbf{U}_{\mathrm{BB}}$ for $i = 1, 2, \ldots, N - 1$.

4.6 Transport and Device Performance Metric Calculation

Having obtained the device TB Hamiltonian, we then employed the NEGF formalism to compute the quantum transport properties of the TIFET [64]. The current through the nanoribbon channel is calculated using the Landauer–Büttiker formalism:

$$I = \frac{q^2}{h} \int_{-\infty}^{\infty} T(E) \left[f_1(E) - f_2(E) \right] dE \tag{4.6}$$

In this equation, $T(E)$ is the transmittance of the electron with energy E, given by:

$$T(E) = \mathrm{Trace}\left[\Gamma_1 G^R \Gamma_2 G^A \right] \tag{4.7}$$

where G^R is the retarded Green's function, which is computed for the TIFET channel using the following equation:

$$G^R = \left[EI - H - \Sigma_1 - \Sigma_2 - U \right]^{-1} \tag{4.8}$$

Here, Σ_1 and Σ_2 are the self-energy matrices for the left and right leads [65], respectively, and $\Gamma_1 = i\left(\Sigma_1 - \Sigma_1^\dagger\right)$ and $\Gamma_2 = i\left(\Sigma_2 - \Sigma_2^\dagger\right)$ are the broadening matrices, which represent the scattering rates from the left and right leads. $f_1(E)$ and $f_2(E)$ are the Fermi–Dirac distribution functions at the source and drain leads, respectively, E is the energy of the electron, and h is Planck's constant. $G^A = G^{R\dagger}$ is the advanced Green's function.

The potential energy U throughout the channel is given by:

$$U = \frac{eV_{ds}x}{L} + \lambda_v \sum_{i,s} \xi_i c_{i,s}^\dagger c_{i,s} \tag{4.9}$$

There are two components in the potential energy U. The first term corresponds to the channel potential due to the drain–source voltage. Here, L is the channel length, and x is the position of the atomic site in the nanoribbon with respect to the source end. We assume that the potential due to the drain–source voltage V_{ds} varies linearly across the channel. The second term represents the staggered sublattice potential, which breaks the inversion symmetry. In this term, $\xi_i = +1\,(-1)$ for sublattice A (B), and $c_{i,s}^\dagger$ ($c_{i,s}$) is the second quantized creation (annihilation) operator with spin $s \in \{\uparrow, \downarrow\}$ at site i. λ_v is the strength of the staggered sublattice potential, which depends on E_z, and is given by:

$$\lambda_v = \alpha_v e E_z \tag{4.10}$$

where α_v is a material-dependent parameter [66]. The out-of-plane electric field in the nanoribbon channel, E_z, is given by [67]:

$$E_Z = \frac{V_{TG} - V_{BG}}{\left(\dfrac{2\,t_{\mathrm{HfO_2}}\,\epsilon_{\mathrm{WTe2}}}{\epsilon_{\mathrm{HfO_2}}} + t_{\mathrm{WTe2}}\right)} \tag{4.11}$$

where V_{TG} and V_{BG} are the top and bottom gate voltages, respectively. We applied the electric field boundary condition at the metal–insulator and insulator–channel interfaces in our double-gate structure to derive the formula for Ez. The proposed device structure operates as a junctionless transistor, since the source, channel, and drain regions are all composed of the same material—WTe_2.

From the drain current characteristics, we get the *subthreshold swing* (SS), a critical performance indicator, representing the efficiency of switching from the off-state to the on-state. SS is calculated numerically from the logarithmic slope of the $I_D - V_{GS}$ curve in the subthreshold regime using:

$$SS = \frac{d\left(\log_{10} I_D\right)}{dV_{GS}} \tag{4.12}$$

A lower SS value implies a sharper switching transition, which is highly desirable for low-power electronic applications.

Furthermore, two important performance metrics related to device speed and energy consumption are the *intrinsic delay time* (τ) and the *dynamic power dissipation* (DPD). These are defined as follows:

$$\tau = R \cdot C_t = \frac{C_t \cdot V_{DS}}{I_{\text{on}}} \tag{4.13}$$

$$DPD = V_{DS} \cdot I_{\text{on}} \cdot \tau \tag{4.14}$$

Both τ and DPD are dependent on the total capacitance of the device (C_t) and the on-current (I_{on}). While the calculation of I_{on} follows from Equation 4.6, the computation of C_t requires a detailed understanding of the device geometry and electrostatics, which is elaborated below.

In our proposed device structure (see Fig. 4.3), the FET is configured as a DGFET, with the channel sandwiched between top and bottom oxide layers. The diffusion capacitance is negligible in this context and thus excluded from consideration. The total capacitance C_t is expressed as:

$$C_t = \left(\frac{1}{C_{\text{ch}}} + \frac{1}{C_{\text{ox_top}}} + \frac{1}{C_{\text{ox_bot}}} \right)^{-1} \tag{4.15}$$

Here, C_{ch} is the channel capacitance (or quantum capacitance), and $C_{\text{ox_top}}$ and $C_{\text{ox_bot}}$ are the top and bottom oxide capacitances, respectively.

The channel (quantum) capacitance, C_{ch}, is proportional to the DOS of the channel material and is calculated using [68]:

$$C_Q = q^2 \int_{-\infty}^{\infty} D(E) \left(\frac{1}{4k_B T} \operatorname{sech}^2\left(\frac{E}{2k_B T} \right) \right) dE \tag{4.16}$$

where e is the elementary charge, k_B is Boltzmann's constant, T is the temperature, and $D(E)$ is the DOS at energy E. This integral effectively averages the DOS within an energy window of a few $k_B T$ around the Fermi level.

The DOS $D(E)$ is computed from the NEGF formalism as:

$$D(E) = \frac{1}{2\pi} \operatorname{Tr}\left[G^R(E) \cdot (\Gamma_L + \Gamma_R) \cdot G^A(E)\right] \tag{4.17}$$

where $G^R(E)$ and $G^A(E)$ are the retarded and advanced Green's functions, respectively, and Γ_L, Γ_R represent the coupling matrices of the source and drain electrodes.

The oxide capacitances are calculated using the standard parallel-plate capacitor formula:

$$C_{\text{ox}} = \frac{\varepsilon_0 \, \varepsilon_{\text{ox}} \, L}{d_{\text{ox}}} \tag{4.18}$$

where ε_0 is the vacuum permittivity, ε_{ox} is the relative dielectric constant of the oxide material, d_{ox} is the thickness of the oxide layer, and L is the length of the device.

All capacitances are calculated per unit width of the device. The total capacitance C_t enables the calculation of τ and DPD via Equations 4.13 and 4.14, thereby providing a comprehensive assessment of the device's speed and power performance.

Scaling is required to compare our results with IRDS data fairly, since IRDS values are reported for reference device dimensions. Metrics like intrinsic delay and dynamic power dissipation depend on device width and length, so scaling normalizes the data and ensures that differences reflect technology performance rather than geometry.

From Equation 4.13, we observe that the delay τ is proportional to the total capacitance C_t. Since C_t is proportional to $W \cdot L$, we can write

$$\tau \propto W \cdot L \tag{4.19}$$

If the device dimensions W and L change, then for the new values W_{new} and L_{new}, the delay scales as

$$\tau_{\text{new}} = \frac{W_{\text{new}} \cdot L_{\text{new}}}{W_{\text{old}} \cdot L_{\text{old}}} \cdot \tau_{\text{old}} \tag{4.20}$$

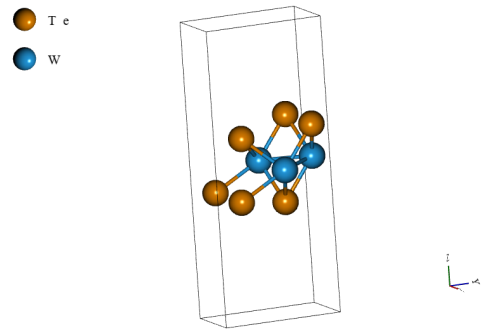
Accordingly, the new DPD can be expressed as

$$DPD_{\text{new}} = \frac{\tau_{\text{new}}}{\tau_{\text{old}}} \cdot DPD_{\text{old}} = \frac{W_{\text{new}} \cdot L_{\text{new}}}{W_{\text{old}} \cdot L_{\text{old}}} \cdot DPD_{\text{old}} \tag{4.21}$$

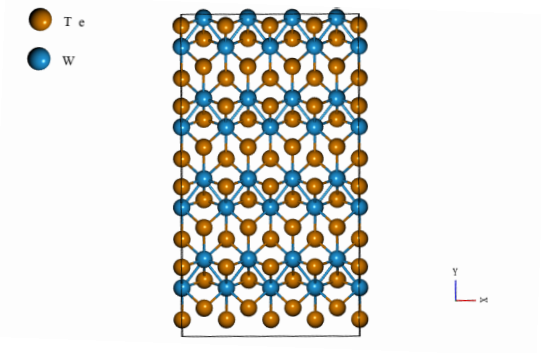
CHAPTER 5: RESULTS AND DISCUSSION

5.1 1T′-WTe_2 monolayer

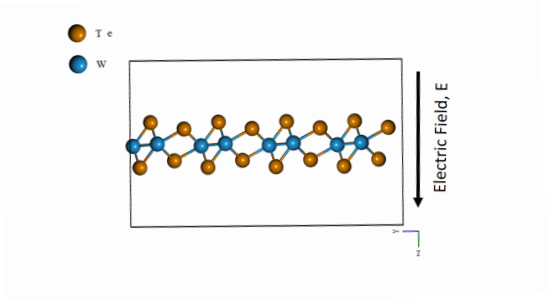
The monolayer of TMDC materials has a 2D crystal structure. It consists of covalently bonded X-M-X sandwiches (M=Mo,W,Ta : X=Te,S,Se). Among them, 2H-WTe2 has periodically hexagonal symmetry similar to graphite. On the other hand, the distorted structure 1T′-WTe_2 has orthorhombic structure. 1T′-WTe_2 is more stable than 1T-WTe_2. In fig. 5.1(a), the unit cell of 1T′-WTe_2 two dimentional(2D) monolayer is shown in different direction. The dark blue one is W and the brown one is Te and the axes are given in x, y, and z directions. In a unit cell, there are two W atoms and four Te atoms. In fig. 5.1(b). and fig. 5.1(c), we have shown the top view and side view of 1T′-WTe_2 monolayer respectively. The direction of the applied electric field is along Z axis as shown in fig. 5.1(c).



(a) Unit cell of 1T′-WTe_2 monolayer



(b) Top view of 1T′-WTe_2 monolayer



(c) Side view of 1T′-WTe_2 monolayer

Figure 5.1: Different views of the 1T′-WTe_2 monolayer.

5.2 Band Fitting

In fig. 5.2, we have shown the bulk bandstructure of 1T′-WTe_2 in {M, Y, Γ, X, M} directions. Now MLWF is utilized to fit the bands around the fermi level. The fitting bandstructure using

MLWF has also been shown in fig. 5.2. The red curve represents the DFT bandstructure and the dotted blue curve represents MLWF bandstructure. In these bands, each tungstan (W) has ten bands and each telluride (Te) has six bands. The total number of bands are 44 bands (4 Te atoms $\times$ 6 bands for each atom = 24 P orbital bands and 10 W atoms $\times$10 bands for each atom = 20 d orbial bands). By fitting the bands from DFT calculation, MLWF produces the tight binding hopping parameters which are required to calculate $\mathbb{Z}_2$ topological invariant.

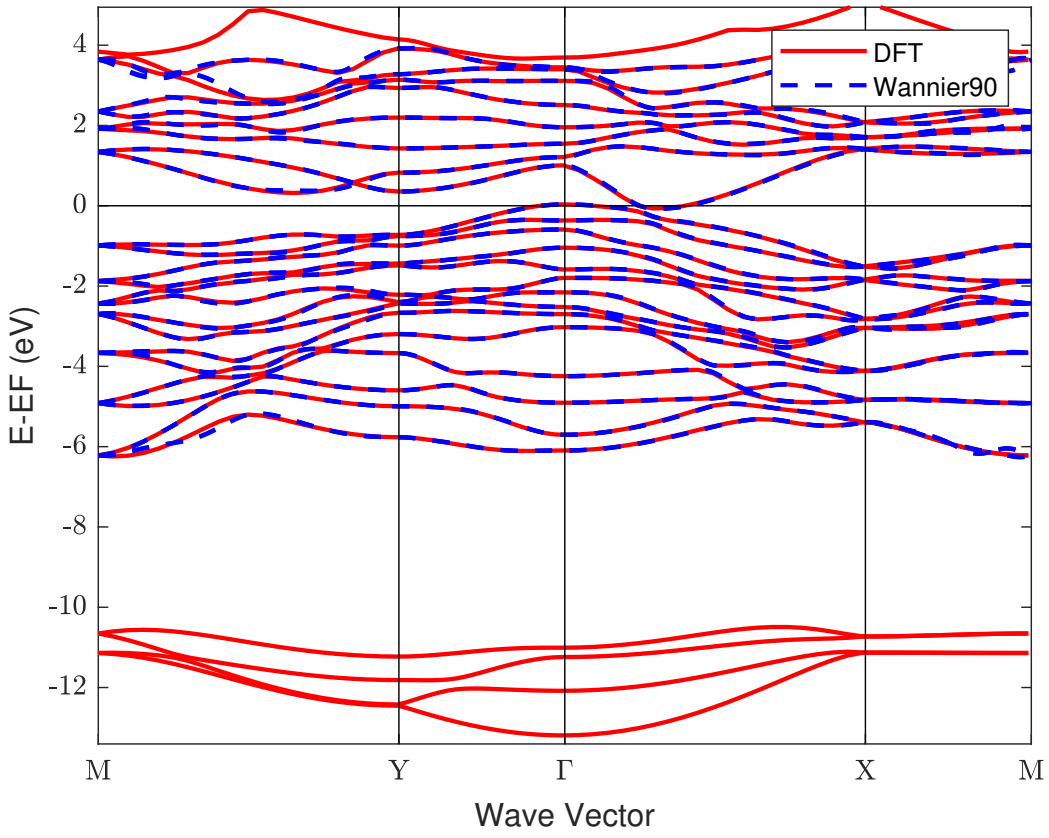
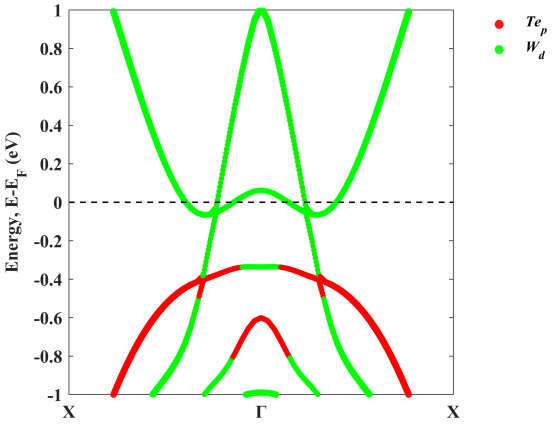


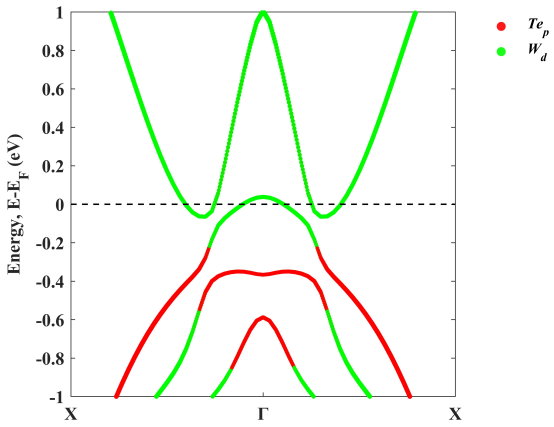
Figure 5.2: Bulk bandstructure of 1T′-WTe_2 using DFT and MLWF

5.2.1 Orbital-projected band structure of 1T′-WTe_2 monolayer

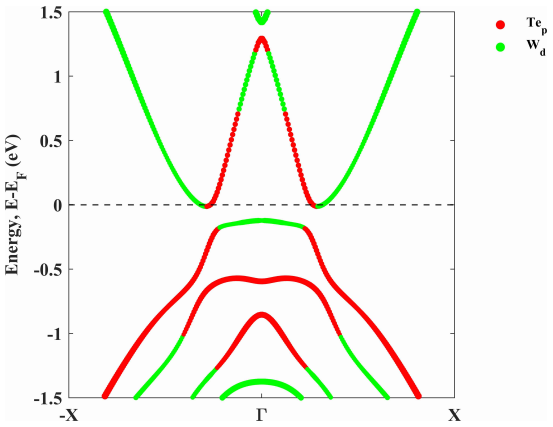
The orbital projected band structure is used to study the role of various atomic orbitals in band structure of a material. The present method also provides a way to understand how the contributions of s, p, and d orbitals from different atoms are taken in each electronic state at different positions in the brillouin zone. They are represented in figures 5.3(a-f) by the red curves for the d orbitals of the W atoms and blue curves for the p orbitals of the Te atoms.



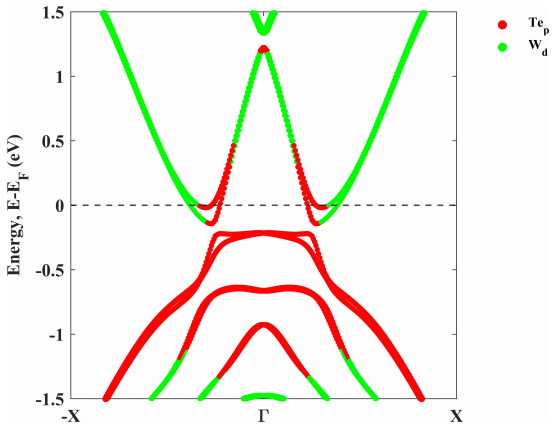
(a) BS without SOC and E=0.0 V/Å, $\mathbb{Z}_2 = 0$



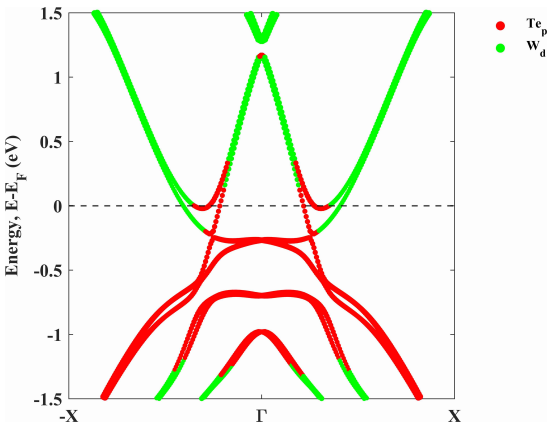
(b) BS with SOC and E=0.0 V/Å, $\mathbb{Z}_2 = 1$



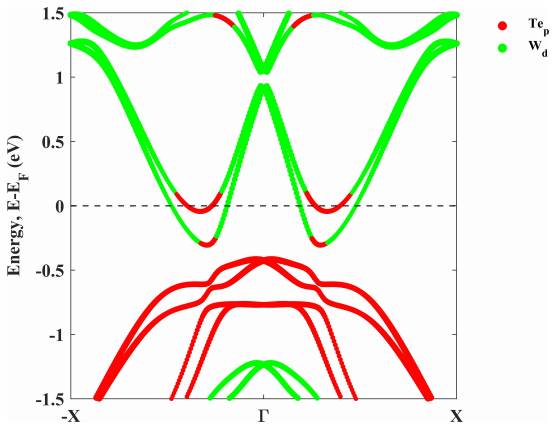
(c) BS with SOC and E=0.0 V/Å, $\mathbb{Z}_2 = 1$



(d) BS with SOC and E=0.0477 V/Å, $\mathbb{Z}_2 = 1$



(e) BS with SOC and E=0.1431 V/Å, $\mathbb{Z}_2 = 1$



(f) BS with SOC and E=0.1908 V/Å, $\mathbb{Z}_2 = 0$

Figure 5.3: Orbital-projected band structures of 1T′-WTe_2 (a) Without SOC using PBE, (b) With SOC using PBE, (c-f) With SOC using HSE06.Green symbols represent d orbital projections, and red symbols represent p orbital projections.

These figures illustrate the orbital projected band structure in {-X, Γ, X} directions for 1T′-WTe_2 monolayer as it is easier to observe the changes of bands under the impact of electric

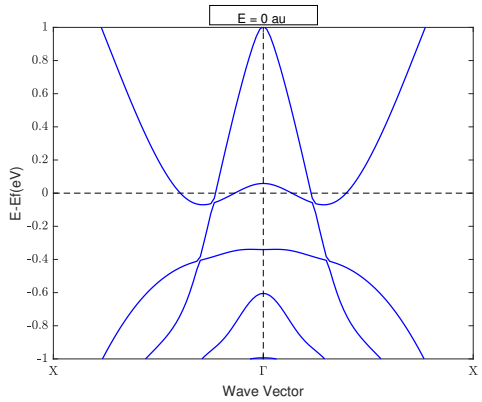
field in these particular directions.

The band structure calculated using PBE without SOC as shown in figure 5.3(a), demonstrates no bandgap whereas the band structure with PBE+SOC, shown in figure 5.3(b), exhibits a negative indirect bandgap. On the other hand, the band structure of WTe_2 calculated by HSE06+SOC and shown in figure. 5.3(c), demonstrates a positive indirect bandgap, approximately 114 meV with the top of the valence band and the bottom of the conduction band being located at gamma and (0.63767, 1.15858) in the k space, respectively. It should be noted that the obtained positive gap does not come from in-plane strains since HSE06+SOC is used for the band structure calculation only. The primary distinction between figures 5.3(a) and 5.3(c) is the transition from the d orbital to the p orbital along Γ to X point. Due to spin-orbit coupling (SOC), some parts of valence band maximum and conduction band minima transition from the d orbital to the p orbital along Γ to X point , indicating band inversion.

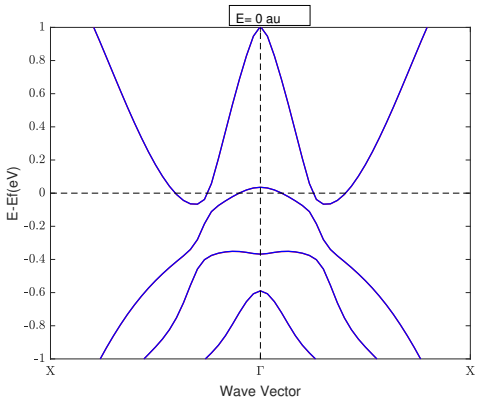
Figures 5.3(d-f) demonstrate the orbital-projected band structures along the -X, Γ, X paths for the 1T′-WTe_2 monolayer after the application of the electric fields. These figures illustrate how increasing the electric field strength affects on the band structure of 1T′-WTe_2. At an electric field of 0.0477 V/Å, the band splitting intensifies, and the band gap decreases, as shown in Fig. 5.3(d) with the topological invariant Z2 =1. However, when the electric field becomes the critical value 0.1431 V/Å, as shown in Fig. 5.3(e), the Z2-number just stays unchanged at Z2=1, and the gap closes. When the electric field exceeds this critical value, the configuration shown in Fig. 3(f) with the field 0.1908 V/Å, exhibits a bandgap opening and the topological number Z2 switches to 0, which implies a material phase transition. This implies a phase transition of the system from a topologically non-trivial to a trivial phase above the critical electric field.

5.2.2 Phase transition of 1T′-WTe_2 bulk monolayer

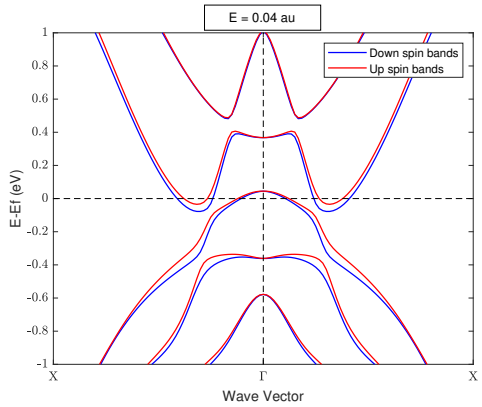
In this section we have discussed the phase transition of 1T′-WTe_2 bulk monolayer under the effect of an applied electric field, using DFT in figures 5.4 (c-f). In these figures, the red curve is for up spin bands and the blue curve is for down spin bands. In figs. 5.4(a,b), as the up spin and down spin bands are degenerate, only blue curves are shown. On the other hand, in the bandstructures of figs. 5.4(c-f), the up spin and down spin bands are not degenerate due to electric field. They start splitting in fig. 5.4(c) where we allocate electric field E =0.04 a.u. (ie, 1 a.u.(atomic unit) = $51.4220632 \times 10^{10}$ V/m). The 1T′-WTe_2 in fig. 5.4(c) still shows topological property as The topological invariant $\mathbb{Z}_2$=1 is computed. In Fig. 5.4(d), the electric field is increased to 0.05 a.u., and it is viewed that the bandgap becomes zero. We increase the electric field to 0.06 a.u. and 0.08 a.u. for fig. 5.4(e) and fig. 5.4(f) respectively. In Fig. 5.4(e), we have found a bandgap around 25.9 mev. On the other hand, the bandgap becomes zero for fig. 5.4(f). The 1T′-WTe_2 presents a trivial insulator with $\mathbb{Z}_2$=0 for figs. 5.4(d,e,f).



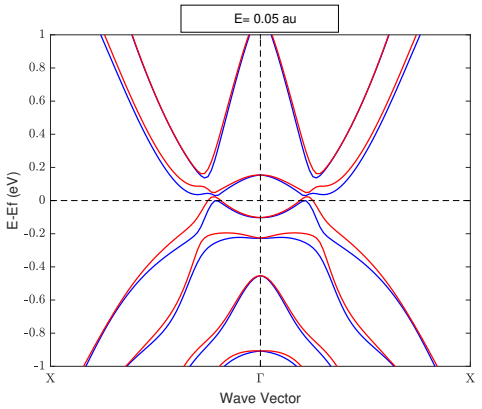
(a) Bandstructure without SOC, $\mathbb{Z}_2$=0



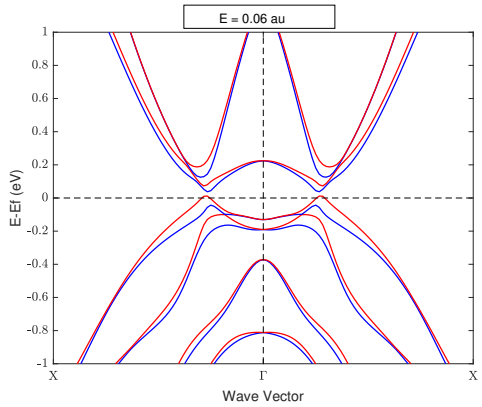
(b) Bandstructure with SOC, $\mathbb{Z}_2$=1



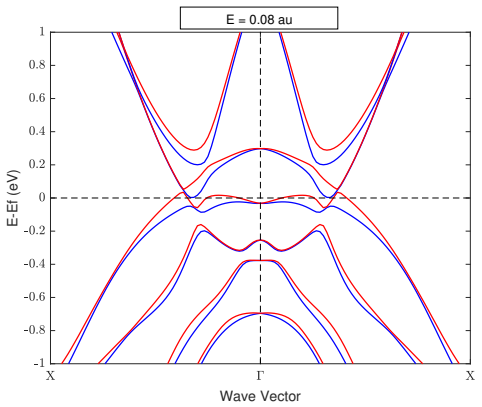
(c) Electric field, E=0.04 au, $\mathbb{Z}_2$=1



(d) Electric field, E=0.05 au, $\mathbb{Z}_2$=0



(e) Electric field, E=0.06 au, $\mathbb{Z}_2$=0

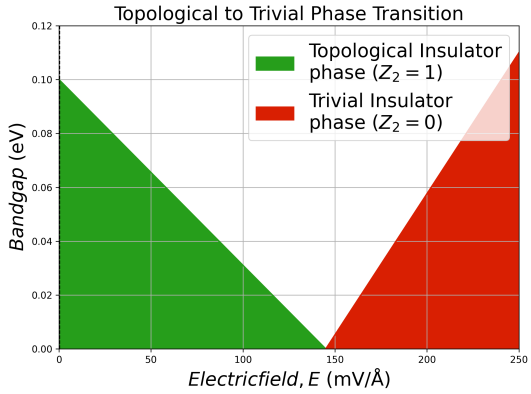


(f) Electric field, E= 0.08au, $\mathbb{Z}_2$=0

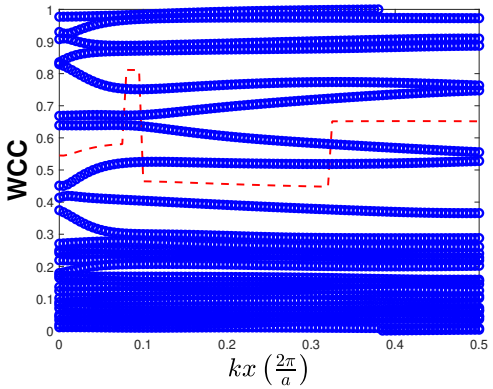
Figure 5.4: Electric field effect on the bandstructure of 1T′-WTe_2. [1 a.u.(atomic unit) = $51.4220632 \times 10^{10}$ V/m]

5.2.3 Topological invarient $\mathbb{Z}_2$

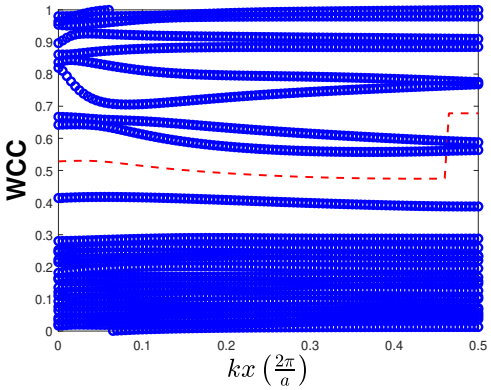
We computed $\mathbb{Z}_2$ for the 1T′-WTe_2 monolayer using MLWF tools to distinguish the type of insulator, taking into account SOC and the applied electric fields. Figure 5.5(a) shows the change in bandgap with respect to the applied electric field. We observe that the bandgap decreases to approximately zero while applying electric field from zero to 0.1431 V/Å. At E=0.1431 V/Å,



(a) Bandgap vs electric field



(b) $\mathbb{Z}_2 = 1$



(c) $\mathbb{Z}_2 = 0$

Figure 5.5: (a) Energy bandgap as a function of electric field. (b) Wannier charge centre (WCC) evolution for $E = 0$ V/Å. (c) WCC evolution for $E = 0.2861$ V/Å.

the electric field reaches a critical value as the bandgap at this electric field is approximately zero. Beyond this point, the bandgap increases with increasing the applied electric field. Both green and red curves of figure 5.5(a) are almost linear. For figure 5.5(a), the results indicate that a significant electric field greater than 0.1431 V/Å triggers the transition from the topological insulator phase to the trivial insulator phase. The topological phase is indicated by the green region, while the red region corresponds to the trivial phase.

In table 5.1, we discuss about the critical electric field and bandgap of different materials. The

Materials	**Bandgap**	**Critical Electric Field**
Si	1.1 eV	300 kV/cm
SiO_2	9 eV	10 MV/cm
Air	-	30 kV/cm
HfO_2	5.5-6 eV	5-10 MV/cm

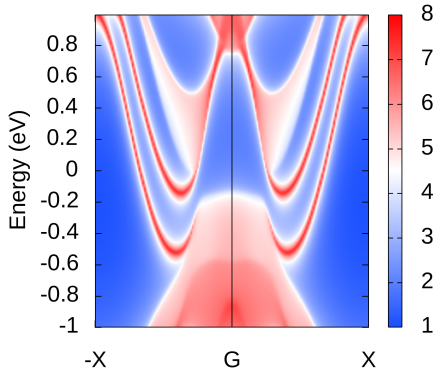
Table 5.1: Bandgap and Critical Electric Field for Various Materials

critical electric field is the minimum electric field strength needed to induce dielectric breakdown or failure in an insulating material. If the electric field exceeds this threshold, the material can no longer resist the electrical stress, leading to the formation of conductive channels and

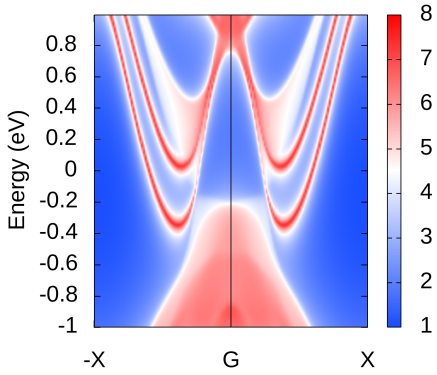
a loss of its insulating properties. The breakdown voltage is the voltage at which dielectric breakdown happens in an insulating material. This is the voltage applied across the material that causes it to lose its insulating abilities and begin to conduct electricity. The breakdown voltage is influenced by the material's thickness and its critical electric field. The critical electric fields of various materials, such as Si, SiO_2, air, and HfO_2, have been documented, but the critical electric field for 1T′-WTe_2 has not yet been reported by any researchers. Our study determines the electric field for phase transition to be 14.3 MV/cm, which is slightly higher than that of the materials listed in Table 5.1. By using a bilayer, trilayer, or multilayer configuration of WTe_2, the critical electric field required to induce a phase transition could potentially be lowered. The addition of layers might alter the material's electronic structure, improving its ability to endure higher electric fields without dielectric breakdown. As the number of layers increases, the overall electronic interactions and screening effects are modified, which can lower the threshold for phase transition. This approach may open new opportunities for developing devices that are more stable and require less power, with phase transitions occurring at lower electric fields. However, further experimental research is needed to fully understand the influence of layer-dependent properties on the critical electric field for phase transition in WTe_2.

Figures 5.5(b) and 5.5(c) present the progression of the Wannier charge center (WCC) in relation to the applied electric field, depicted in figure 5.5(b) for E = 0 V/Å and figure 5.5(c) for E=0.2861 V/Å. The figures illustrate the Wannier charge center (WCC) lines represented by blue curves, with the red dotted curve indicating the largest gap between the WCCs. The determination of the $\mathbb{Z}_2$ invariant depends on whether the red dotted line intersects the WCC curves an odd or even number of times. An even number signifies a trivial insulator, whereas an odd number denotes a topological insulator. The analysis of 1T′-WTe_2 reveals that in fig.5.5(b), the red curve intersects the WCC curves at nine distinct points, leading to a determination of $\mathbb{Z}_2$=1 which indicates a topological insulator. In contrast, for the 1T′-WTe_2 with E = 0.2861 V/Å in figure 5.5(c), the red dotted curve intersects the WCC curves twice, resulting in $\mathbb{Z}_2$ = 2 mod 2 = 0, which signifies a trivial insulator.

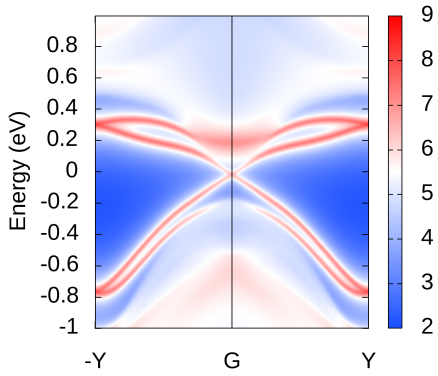
Figures 5.6(a-d) demonstrate the surface states, where we utilize the surface green's function technique to calculate the surface state of a semi slab system. Figures 5.6(a,c) show non-trivial edge states as they connect valance band and conduction band together. These results correspond to the case without an applied electric field. However, when an electric field, E=0.247 V/Å is applied as shown in figures 5.6(b) and 5.6(d), we observe that the edge states disappear around the Fermi level, creating a gap between the valance band and conduction band. This behavior indicates the signature of trivial insulator. The above analysis suggests that 1T′-WTe_2 changes its phase from topological to trivial after a certain electric field.



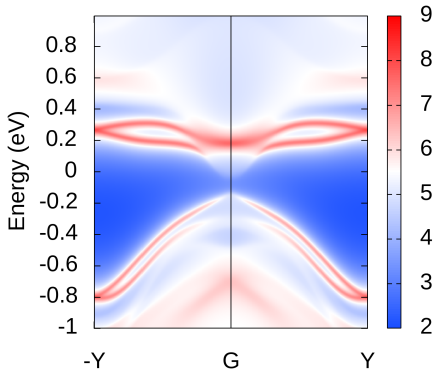
(a) $\mathbb{Z}_2$=1



(b) $\mathbb{Z}_2$=0



(c) $\mathbb{Z}_2$=1



(d) $\mathbb{Z}_2$=0

Figure 5.6: Surface states vs electric field. The electric field (a,c) E = 0 V/Å and (b,d) E = 0.2861 V/Å

5.2.4 Effect of an applied electric field on pristine zigzag and armchair 1T′-WTe_2 nanoribbon

In this section, we examine the impact of the applied electric field on both pristine zigzag and armchair 1T′-WTe_2 nanoribbons. For our case, we simulated zigzag (126 atoms in a unit cell , about 13.3103nm) and armchair (126 atoms in a unit cell, about 7.3455nm) nanoribbons. Figures 5.7(a) and 5.7(d) illustrate the electronic band structure of zigzag and armchair 1T′-WTe_2 nanoribbons, respectively. we observe the edge states around the Fermi level for both nanoribbons. The edge states are denoted by the red curves, whereas the bulk states are illustrated by the blue curves. The valence and conduction bands display a distinctive characteristic of gapless edge states, intersecting linearly at either the Γ or X point for both nanoribbons. To examine the impact of electric field, we implemented a linear electric field along the z axis of the nanoribbons. The electric field influences the onsite potential between the upper and lower layers of Te atoms, resulting in the inequivalence between the two sublattices and thereby breaking

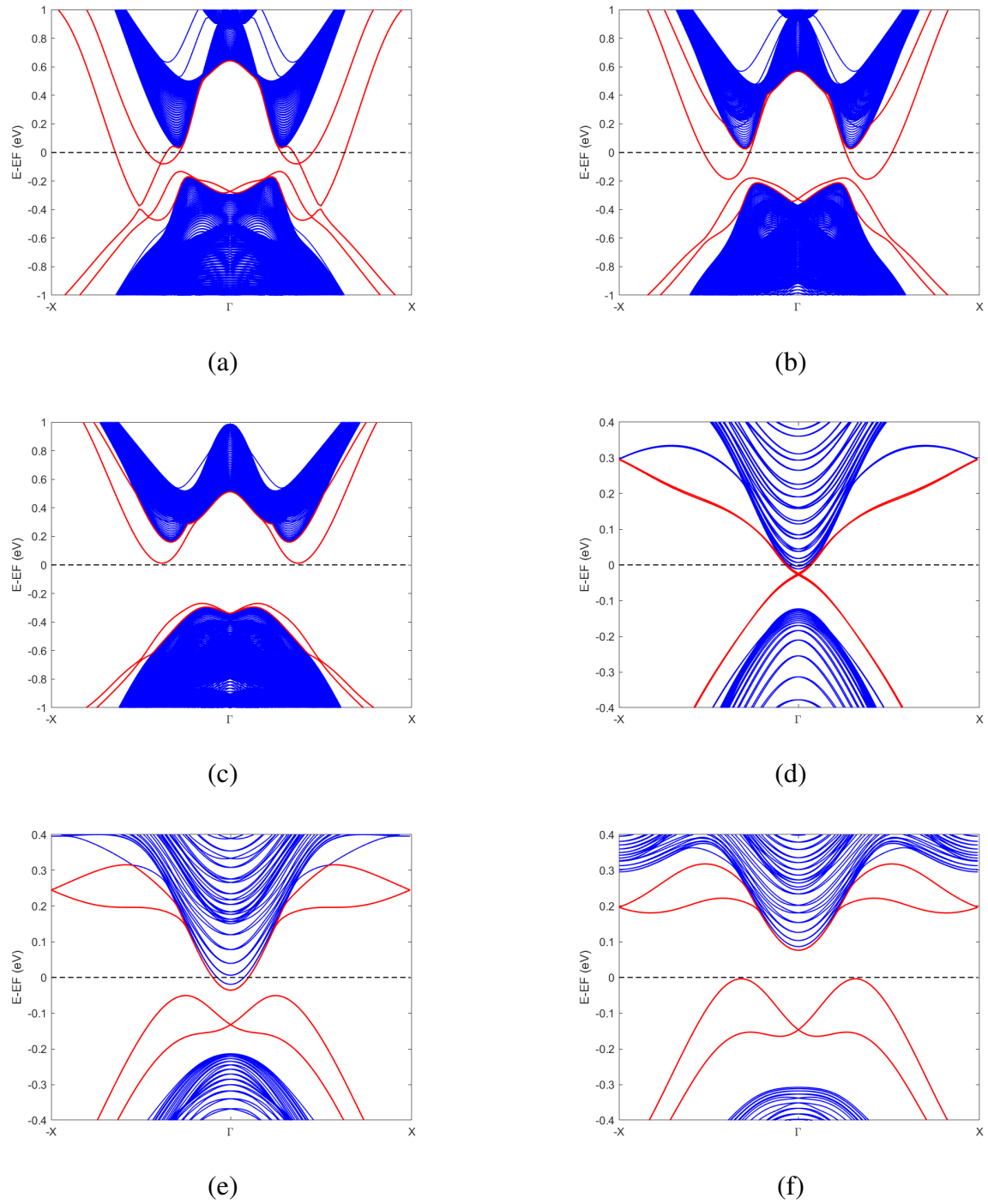


Figure 5.7: Band structures of zigzag nanoribbon of1T′-WTe_2 With SOC-HSE06: (a) without electric field, (b) with E=0.2861 V/Å, (c) with E=0.4769 V/Å, Band structures of armchair nanoribbon of 1T′-WTe_2 With SOC-HSE06 (d) without electric field, (e) with E=0.2861 V/Å, (f) with E=0.4769 V/Å Blue curves represent up spin bands, and red curves represent down spin bands.

the inversion symmetry.

Figure 5.7, subfigures (b), (c), (e), and (f) depict the impact of an external electric field on the band structure of 1T′-WTe_2 nanoribbons. In figure 5.7(b), E = 0.2861 V/Å is applied to

the zigzag nanoribbon; the topologically protected edge states remain visible, although the gap begins to narrow as a result of the external field. The band dispersion is modified in comparison to the zero-field scenario. Furthermore, in figure 5.7(c), we apply an electric field of E = 0.4769 V/Å to the zigzag nanoribbon, resulting in a significant suppression or complete gapping of the edge states; it indicates that the system has experienced a topological phase transition to a trivial insulating state. In figure 5.7(e), an electric field of E = 0.2861 V/Å is applied to the armchair nanoribbon, resulting in an increased band gap. The electric field disrupts inversion symmetry, hence removing degeneracy. In figure 5.7(f), we apply an electric field of E=0.4769 V/Å respectively to the armchair nanoribbon, resulting in splitting of the bands. The edge states start opening and a full bandgap is formed due to the electric field for both nanoribbons. As a result, both nanoribbons lose topological insulator properties.

Figure 5.8 provides the schematic diagram of the band structure of trivial and topological materials. This figure is meant to show the on and off conditions of the material in the channel material application. In the case of topological insulator, topological materials possess special characteristics that render them applicable in application as a channel in electronic devices. The material in this state can conduct current along its surface states and the bulk is insulating hence the conduction of current can be conducted in a very controlled manner. On the application of an electric field, the material may change into a trivial state of insulator, in which the surface states are repressed and the material can no longer carry current. Topological materials are an exciting prospective technology in the creation of advanced electronic devices due to this capability to switch between conducting and insulating states depending on external conditions such as electric fields, such as switches and transistors that can effectively and efficiently operate with low power consumption.

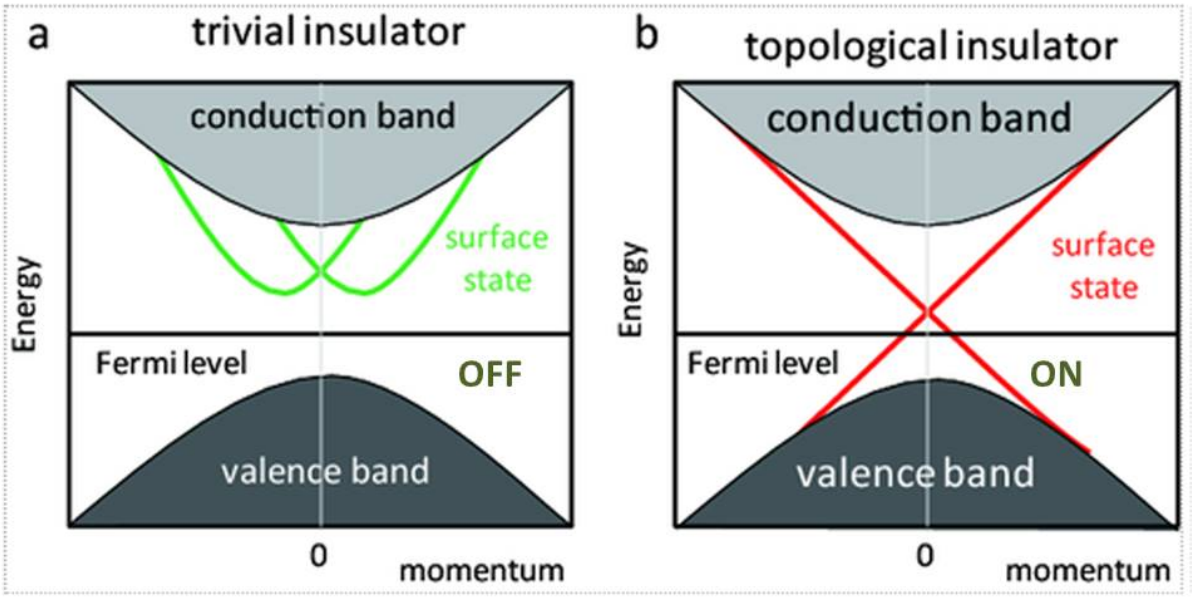
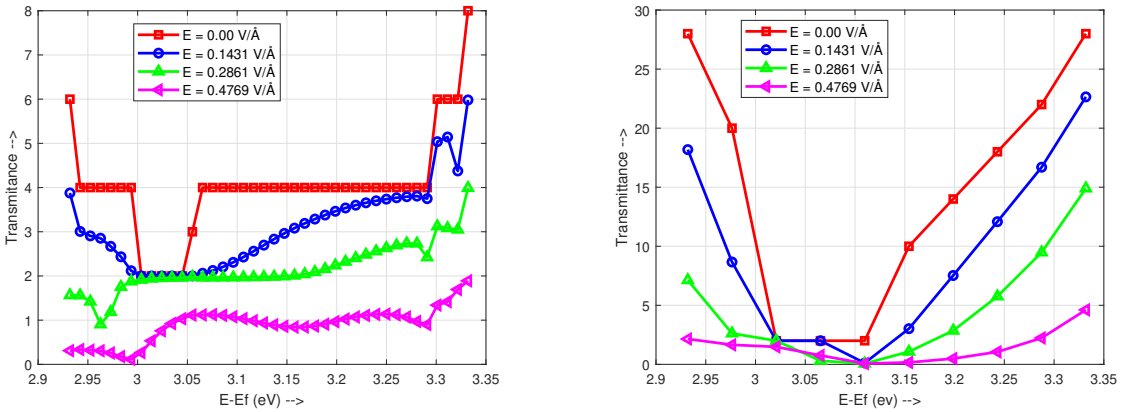


Figure 5.8: Band Structure Of trivial and topological insulator(Prof. Shoucheng Zhang,Stanford University, Stanford, US)

5.2.5 Field-Tunable Transport and FET Application of 1T′-WTe_2 Nanoribbons



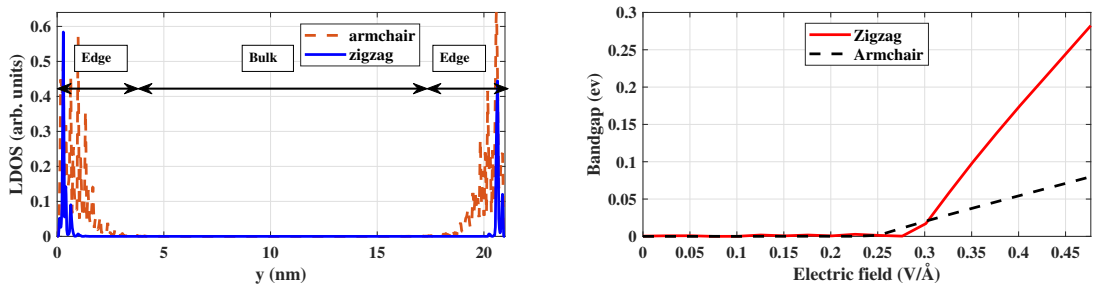
(a) (b)

Figure 5.9: Transmittance vs Energy for (a) Zigzag (b) Armchair nanoribbon

Figures 5.9(a, b) show the transmittance as a function of energy, E for two types of 1T′-WTe_2 nanoribbons. They show the variation of transmittance at different values of the applied elec-

tric fields in a sequcnce: E=0.00 V/Å,0.1431 V/Å,0.2861 V/Å,0.4769 V/Å. Step-like features are observed for transmittance profiles for E=0V/Å indicating the signature of quantised conduction channels. As the electric field further increases, the transmittance decreases and the profile becomes more flat,which indicates the reduction of conductance and probably with the opening of a band gap. It is also noticeable that transmittance remains T=2 until the electric field, E=0.2861 V/Å in fig 5.9(a). In both fig. 5.9(a) and 5.9(b), we observe the edge transport with transmittance , T=2 from energy 3.0ev to 3.05ev approximately for E=0.00 V/Å. As transmittance decreases with increasing electric field, this phenomenon can be exploited in field-effect transistor (FET) applications. The fundamental principle of an FET is to switch it from its on-state to off-state upon the application of a certain electric field. In this material, transmittance becomes nearly zero after applying a specific electric field, leading to negligible current through the channel. As a result, 1T′-WTe_2 nanoribbons are a potential candidate for use as channel materials in FETs.

5.2.6 Electron density profile of zigzag 1T′-WTe_2 nanoribbon



(a) (b)

Figure 5.10: (a)Electron density profile of 1T′-WTe_2 nanoribbon(b)Bandgap of 1T′-WTe_2 nanoribbons with respect to the applied electric fields

In Figure 5.10(a), we present the electron density profile (LDOS) of 1T′-WTe_2 for two different types of nanoribbons. The x-axis indicates the position along the y-direction (in nm) of the nanoribbon, and the LDOS is plotted on the y-axis. The blue dashed line corresponds to the electron density of the zigzag nanoribbon, while the red dashed curve represents the electron density of the armchair nanoribbon. It can be observed from the figure that the LDOS near the edges (within the first 5 nm) of the nanoribbons is much higher than in the bulk regions. It is also worth noting that the edge states of zigzag nanoribbons are more prominent, as sharper peaks appear in the LDOS compared to the peaks of armchair nanoribbon. In the bulk region, the LDOS is almost zero, which further confirms the existence of localized states at the edges, typically associated with topologically protected edge modes in nanoribbons. Figure 5.10(b)

illustrates the correlation between the bandgap (Eg) and the applied electric field for zigzag and armchair 1T′-WTe_2 nanoribbons. Significantly, a zero bandgap is detected in armchair nanoribbons at an electric field of 0.2509 V/Å and in zigzag nanoribbons at 0.27609 V/Å. As the value of the applied electric field increases, the bandgap for both types of nanoribbons also increases. It is shown that when a fixed electric field beyond 0.45 V/Å is applied, the induced bandgap for the zigzag nanoribbon is roughly 0.2024 eV greater than that of the armchair nanoribbon.

5.3 Transfer characteristic of the proposed TFET

Up to now, we have discussed the electronic properties, transport, and robustness of the 1T′-WTe_2 nanoribbon. Now, we will discuss the transfer characteristics of our proposed TIFET in detail. In our TIFET structure, different voltages will be applied to the top and bottom gates, inducing an electric field. When this electric field exceeds E=0.1431 V/Å, a topological phase transition will occur along with bandgap opening due to inversion symmetry breaking.

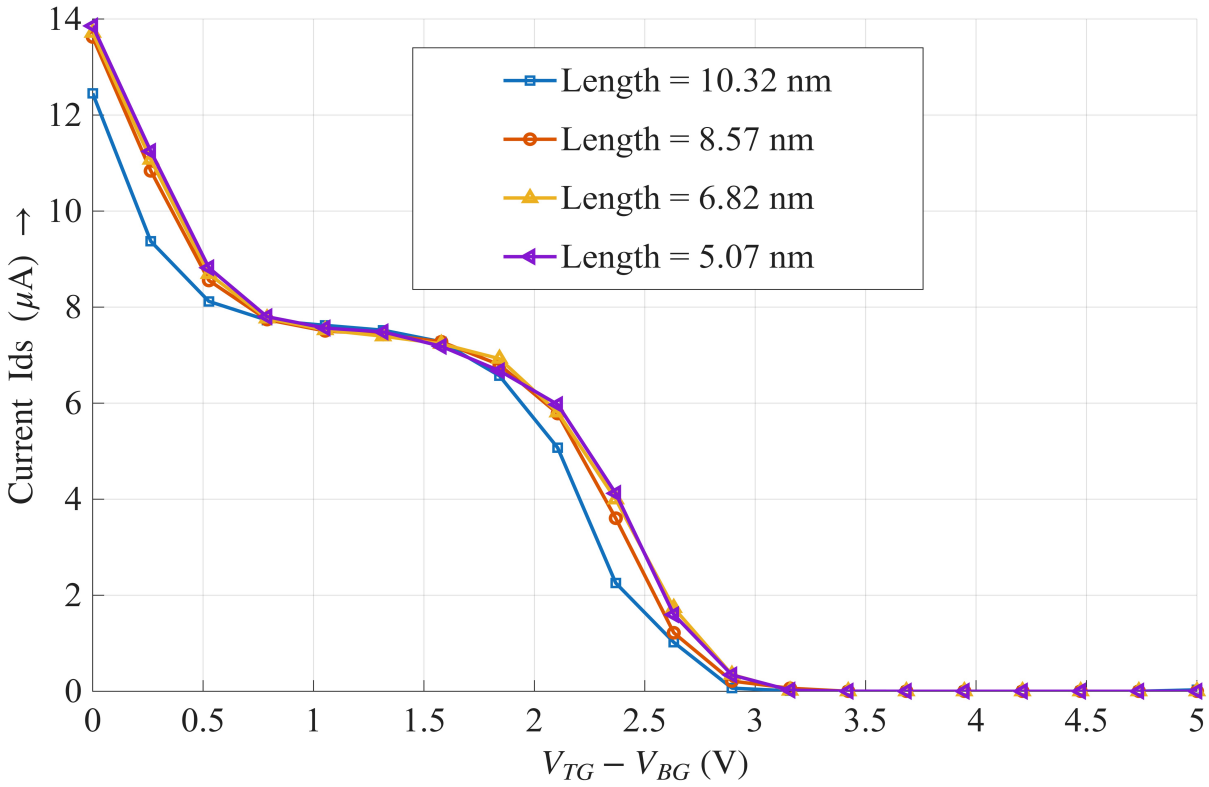


Figure 5.11: Linear transfer characteristics for different channel lengths

Now we will discuss the transfer characteristics of our proposed TIFET. Fig. 5.11 depicts the linear characteristic of our proposed TIFET. The X-axis is labeled as $V_{\mathrm{TG}} - V_{\mathrm{BG}}$ because different voltages were applied to the top and bottom gates to intentionally break inversion symmetry across the device. Unlike conventional MOSFETs that rely on the formation of an

inversion layer, the operation of TIFETs is governed by symmetry-protected edge states. By creating an electric field through asymmetric gating, we induce a topological phase transition by disrupting the inversion symmetry of the QSH insulator nanoribbon. The drain-to-source voltage Vds is set to 0.05 V. The channel length is set to 10.32 nm. The width of nanoribbon channel is approximately 7.6 nm. It is noted that current in QSH insulator channel does not depend on the width of the channel, as electrons only travel through the topologically protected edges of the channel.The channel current at $V_{\mathrm{TG}} - V_{\mathrm{BG}}$ = 0V , for all length is approximately ($I_{DS} = 1.3625 \times 10^{-5}$ A). There are only two modes for electron transport at the two edges of the edge-passivated nanoribbon, which corresponds to two quantum of resistance (25.9 kΩ) in parallel, yielding a resistance value of 13 kΩ.

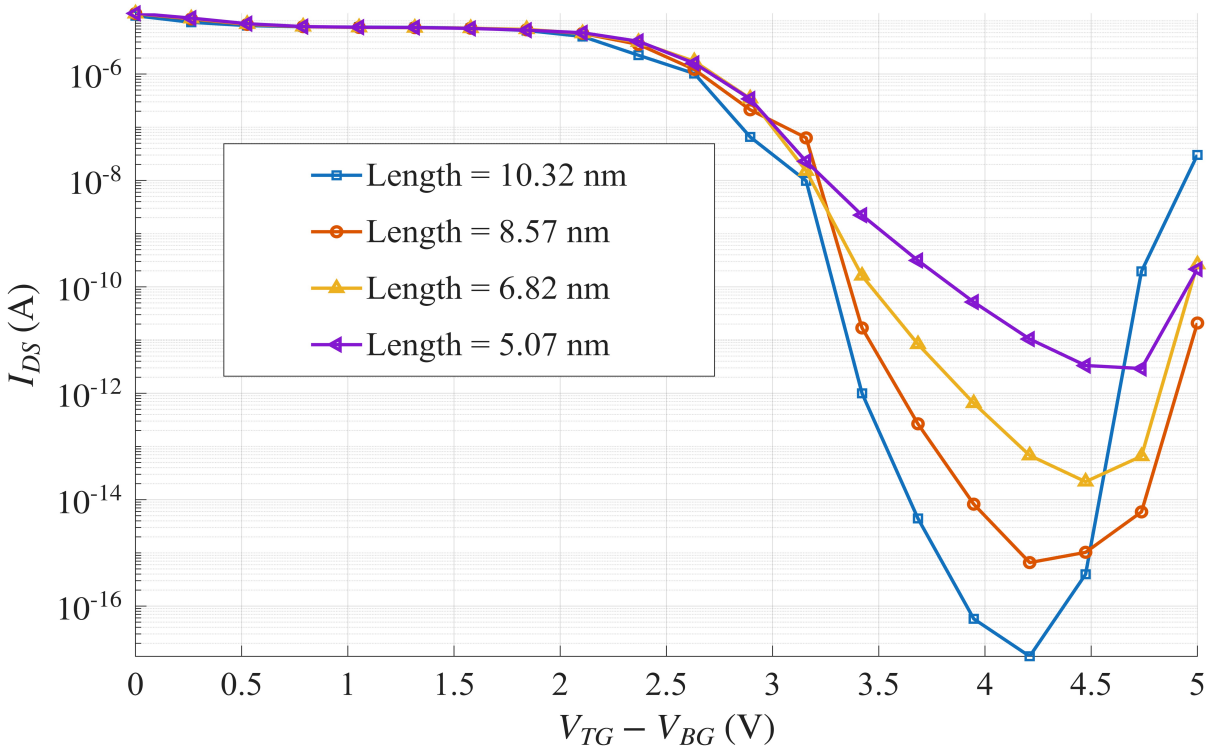


Figure 5.12: Logarithmic transfer characteristics for different channel lengths

The logarithmic transfer characteristic in Fig. 5.12 shows drain current I_D changes with the differential gate bias ($V_{TG} - V_{BG}$) for devices with channel lengths between 5.07 nm and 10.32 nm. At low gate bias, the current stays in the 10^{-6} A range for all channel lengths. Then, as increasing gate bias voltage starts creating bandgap, I_D drops quickly. Each curve has a unique minimum current point, after which the current goes back up because the bandgap starts decreasing with the increase of gate voltage. There is a clear link between channel length and electrostatic control since longer channels have stronger electrostatic control, which leads to much lower minimum leakage currents. The 10.32 nm device has the lowest off-state current, which is close to 10^{-16} A. The 5.07 nm device, on the other hand, has a much higher leakage current, which is around 10^{-11} A, because of short-channel effects.

Although the ON current for our proposed TIFET remains constant with increasing channel length, the OFF current reduces significantly with channel length. This is because, during the OFF state, there is tunneling current between the QSH insulator source and drain through the trivial channel barrier. We know the tunneling probability $T \propto e^{-\alpha L}$, where L is the barrier thickness; as a result, the OFF current reduces significantly with increasing channel length.

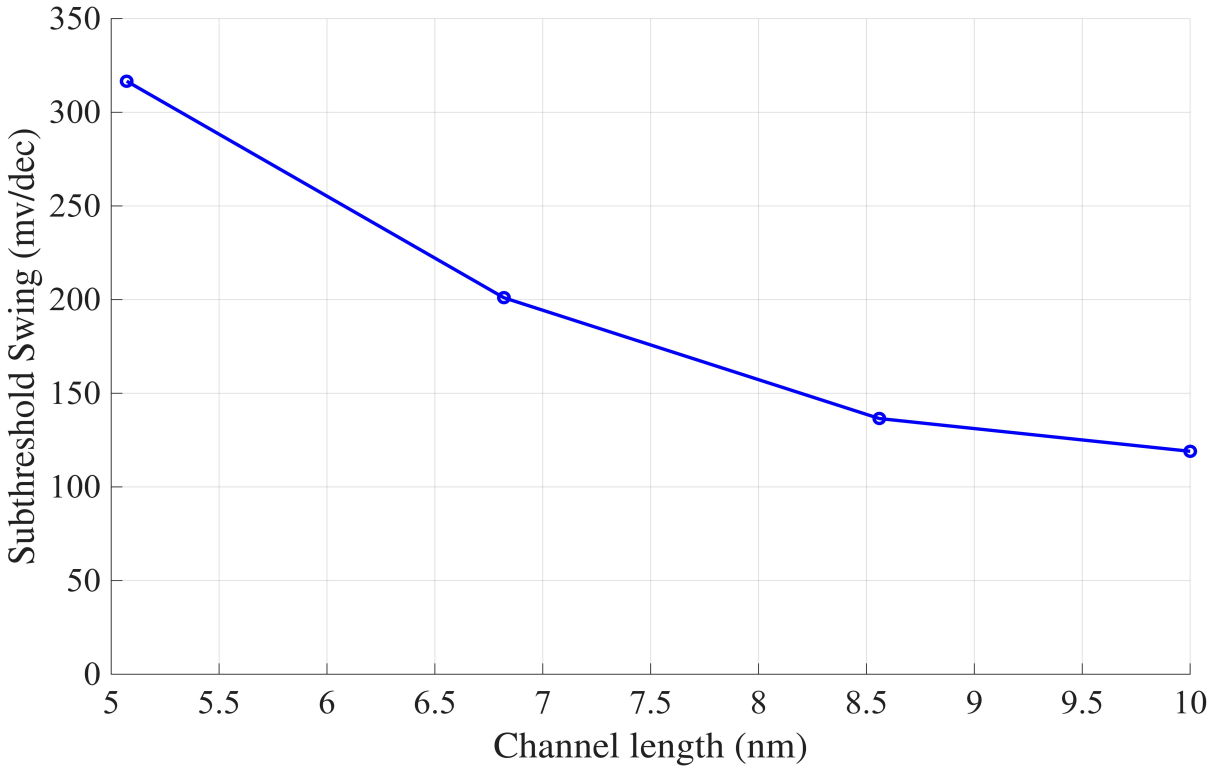


Figure 5.13: Sub-threshold swing for different channel length

Now, if we always keep $V_{BG} = -1.84\,\text{V}$, $V_{TG} = 0\,\text{V}$ will correspond to the ON state, and $V_{TG} = 1.3178\,\text{V}$ will correspond to the OFF state. By this definition, the threshold voltage for our proposed TFET is 1.3178 V. We have also calculated the subthreshold swing (SS), one of the most important figures of merit for transistor operation. The SS is a direct measure of how efficiently the FET switches between the OFF and ON states.

$$SS = \frac{d(V_{TG} - V_{BG})}{d(\log_{10} I_{DS})} \times 1000$$

Figure 5.13 demonstrates the sub-threshold swing with respect to channel length. In figure 5.11, it is observed that on current remains constant with channel length due to ballistic transport in topological edge states. Another important observation is that increasing the channel length reduces the SS. The calculated SS for channel lengths of 5.072 nm, 6.820 nm, 8.569 nm, and 10.318 nm are 316.59 mV/decade, 201.00 mV/decade, 136.50 mV/decade, and 119.01 mV/decade, respectively. Thus, modeling a TIFET with a channel length in the μm range will significantly reduce the subthreshold swing, which is highly desirable for FET performance.

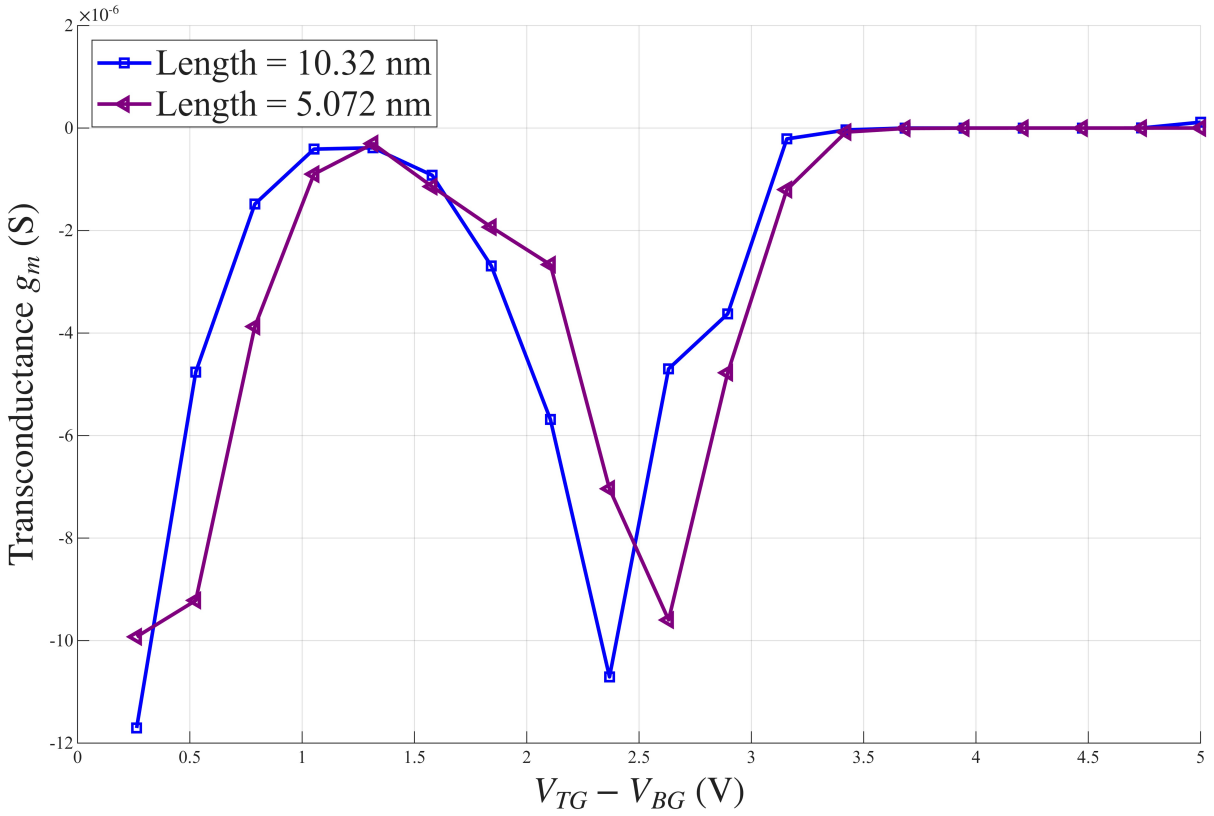


Figure 5.14: Transconductance as a function of $V_{TG} - V_{BG}$

We have also analyzed the transconductance, $g_m = \frac{dI_{DS}}{dV_{GS}}$, of our proposed TFET, as shown in Fig. 5.14. Since the edge states are suppressed by the applied gate voltage, g_m exhibits negative values during the transition between the ON and OFF states. The maximum negative transconductance is observed at 2.3684 V and 2.631 V for channel length 10.32 nm and 5.072 nm , with values of 1.0709×10^{-5} S and 9.59×10^{-6} S, respectively. The experimentally determined carrier mobility of 1T′-WTe_2 is $5000 - 10000\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$ [69].

Up to now, we have discussed the transfer characteristics of the ideal TIFET channel, which means a channel with no defects. However, the existence of defects, impurities, and vacancies is very common in real devices. For this reason, we have also investigated the effect of vacancies on the transfer characteristics of our TIFET as shown in Fig. 5 .15. The 1T′-WTe_2 zigzag nanoribbon TIFET exhibits robustness under vacancy concentrations as large as 2%, though the on current slightly decreases to 8 μA. There is no effect on threshold voltage. However, subthreshold swing increases with the increase in the percentage of defects. FET channels made of trivial materials like graphene are highly sensitive to vacancies. Vacancies act as barriers in graphene nanoribbons, causing the electron wave function to be backscattered, which eventually reduces the ON current ([70]- [71]). Backscattering is prohibited in the TIFET due to topological protection in the edge states provided by time-reversal symmetry. As a result, there will be less heating in the TIFET compared to conventional FETs, making the TIFET more energy-efficient and thermal noise-free.

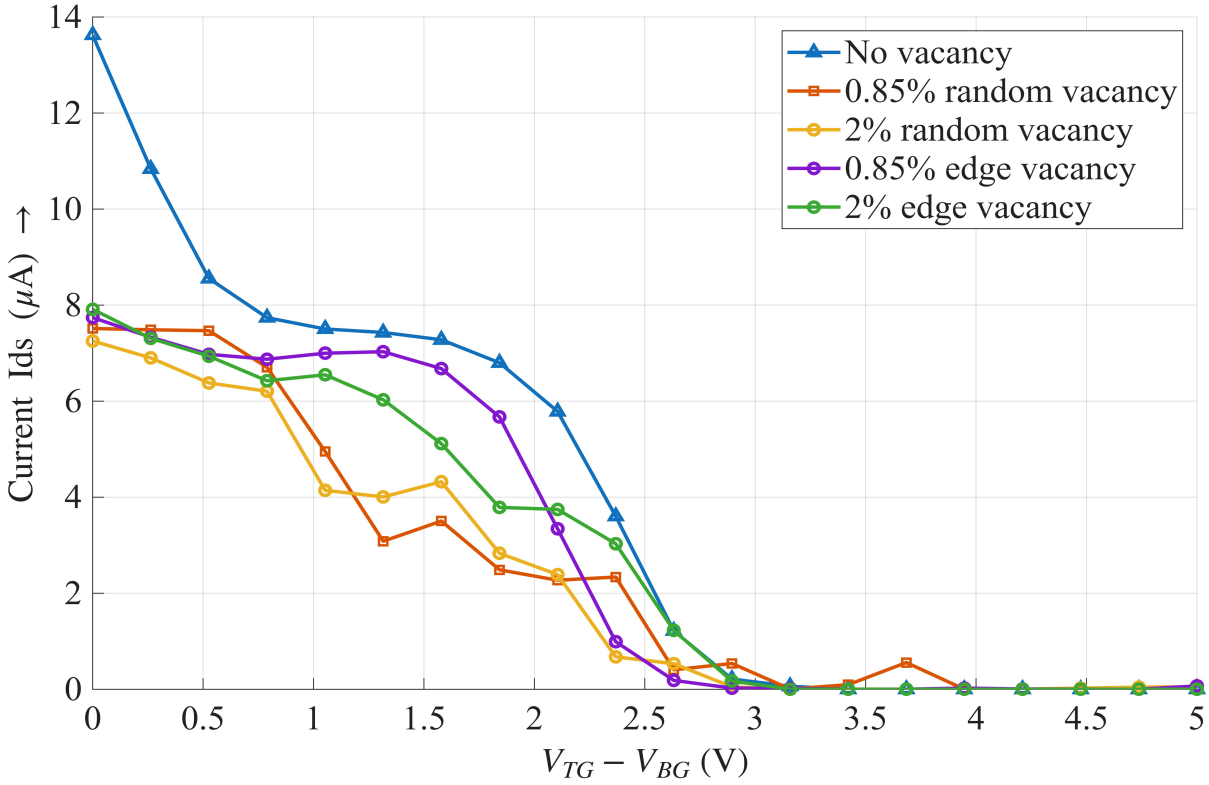


Figure 5.15: Transfer characteristics of TIFET under varying vacancy concentration

5.4 Performance Analysis of the Zigzag 1T′-WTe_2 Nanoribbon TIFET

In this section, we will analyze the performance of our proposed dual-gate zigzag 1T′-WTe_2 nanoribbon TIFET. We begin our analysis by calculating the total capacitance of the TIFET. Capacitance is a very important figure of merit for any field-effect transistor-based device. In an FET, several capacitances contribute to its overall behavior. The intrinsic channel-related capacitances include the oxide capacitance, quantum capacitance, and depletion capacitance. The extrinsic or parasitic capacitances include the overlap capacitance, fringing capacitance, source/drain junction capacitances, gate-to-source capacitance, gate-to-drain capacitance, and gate-to-body capacitance. The SS is reduced when the oxide and quantum capacitances are large, while the parasitic and depletion capacitances are small. In our simulations, only the insulator capacitance and quantum capacitance have been considered. We calculated the insulator capacitance of the HfO_2 layer using the equation for a parallel plate capacitor:

$$C_{HfO_2} = \frac{\epsilon_{HfO_2}}{t_{HfO_2}} \tag{5.4.1}$$

Since two insulator capacitances are in series, the net insulator capacitance is

$$C_i = \frac{C_{HfO_2}}{2} \tag{5.4.2}$$

The quantum capacitance C_Q of the channel will be in series with the insulator capacitance. The total capacitance of our proposed TIFET is:

$$C_{tot} = \frac{C_i C_Q}{C_i + C_Q} \tag{5.4.3}$$

For calculating the quantum capacitance, we used the following formula [72]:

$$C_Q = q^2 \int_{-\infty}^{\infty} D(E) \left(\frac{1}{4k_B T} \text{sech}^2 \left(\frac{E}{2k_B T} \right) \right) dE \tag{5.4.4}$$

where $D(E)$ is the density of states. C_Q is proportional to the density of states averaged over a few $k_B T$ around the Fermi level.

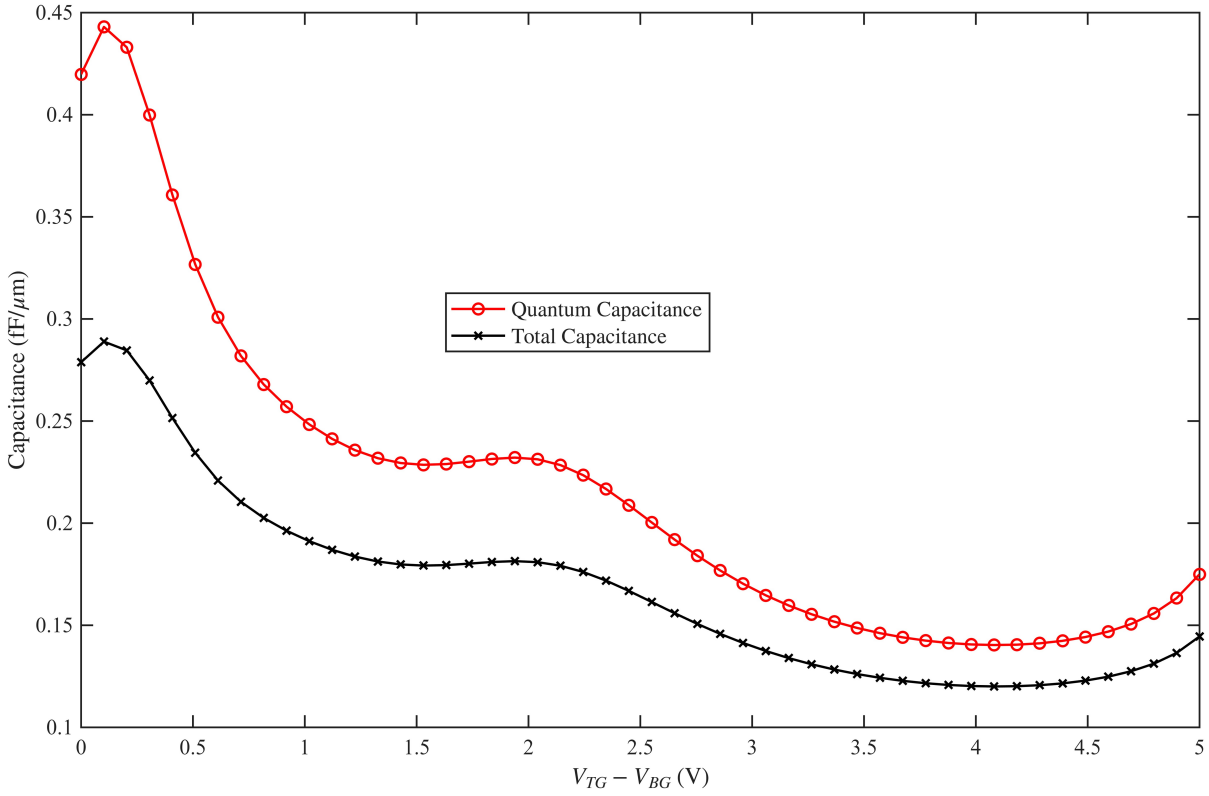


Figure 5.16: Dependence of quantum and total capacitance per unit length on the gate voltage difference ($V_{TG} - V_{BG}$).

The dependency of C_Q, and C_{tot} on the gate voltage is shown in Fig. 5.16. The insulator capacitance (1.6608 fF/μm) does not depend on gate voltage. However, since quantum capacitance is related to the density of states and the application of the electric field opens an energy gap across the Fermi level, the density of states reduces, which eventually reduces the quantum ca-

pacitance. At $V_{TG} - V_{BG} = 0$ V, the quantum and total capacitance values are 0.41977 fF/μm and 0.2788 fF/μm, respectively. At $V_{TG} - V_{BG} = 5$ V, the quantum and total capacitance values reduce to 0.1749 fF/μm and 0.1443 fF/μm, respectively. However, it is observed that the quantum capacitance reached its peak (0.4430 fF/μm) at $V_{TG} - V_{BG} = 0.1024$ V. The quantum capacitance starts decreasing after this voltage.

After calculating the total capacitance (C_{tot}) of the proposed TIFET, we determined the delay time using the expression

$$\tau_d = \frac{C_{tot} V_{dd}}{I_{on}},$$

where V_{dd} is the supply voltage (drain-to-source voltage). The switching delay time represents the duration required for the device to transition between logic states during operation. The calculated delay time is 0.0069 ps. It should be noted that parasitic and depletion capacitances have been neglected in this study; consequently, the reported switching delay values represent an idealized lower bound rather than the actual delay expected in a fabricated device.

We also evaluated the dynamic energy dissipation (DED) per switching event using the relation

$$DED = V_{dd} I_{on} \tau_d.$$

DED refers to the energy consumed during the charging and discharging of the load capacitance when a FET switches between logic states. The resulting value is

$$DED = 4.3448 \times 10^{-6} \text{ fJ per switching event.}$$

This remarkably low DED can be attributed to the highly scaled geometry of the proposed edge passivated zigzag nanoribbon TIFET, which features a width of 7.6 nm and a length of 10.31 nm. Moreover, since parasitic capacitances are neglected, the calculated DED represents an idealized lower bound. It is essential to point out that in the context of dynamic energy-dissipation in FETs, this phenomenonhas no direct relation with that of energy-time uncertainty as constrained by Heisenberg uncertainty principle. The uncertainty principle, represented by $\Delta E \Delta t \geq \hbar$, typically applies to quantum states and their evolution, especially in terms of the limitations on measuring energy over time. However, when taking into account the energy dissipation in FETs due to their switching operation, what matters more is the electrostatic factors such as capacitance and voltage rather than quantum mechanical uncertainty-related quantities. Dynamic energy dissipation (DED) is calculated as the difference between the energization and deenergization of capacitance and may be minimized by adiabatic switching or minimizing capacitance in the equipment. The DED is proportional to the time delay; the higher the time delay, the higher the DED will be. The Heisenberg uncertainty principle does explain though that the more energy you have the less time you have to have it. Therefore, although the Heisenberg uncertainty principle describes a fundamental lower bound in quantum systems, it

is not fundamentally limiting to the energy dissipation behavior of conventional FET switching. Rather, the actual bottlenecks to energy dissipation are due to thermal noise, device efficiency, and intrinsic circuit parameters such asresistance and capacitance.

The dynamic power dissipation (DPD) was calculated using the expression

$$DPD = \frac{1}{2} C_{tot} V_{dd}^2 f.$$

The resulting value is $DPD = 2.1914 \times 10^{-12}$ W for $f = 1$ GHz. The static power dissipation (SPD) during the OFF state is given by $SPD = V_{dd} I_{off}$, yielding a value of $SPD = 4.94 \times 10^{-14}$ W.

Furthermore, the calculated on/off current ratio is 1.26×10^4. The off-state current is $I_{off} = 0.9882\ nA$, while the on-state current is $I_{on} = 12.452\ \mu A$. The subthreshold swing (SS) of the proposed TIFET is remarkably 119.01 mV/decade. A comprehensive comparison of the performance metrics of the proposed TIFET with previously reported devices is presented in Table 5.2.

Table 5.2: Comparison of 1T′-WTe_2 TIFET with other reported devices.

Parameters	**1T′-WTe_2**	**IRDS 2023 [73]**	**IRDS 2025 [74]**	**MoS_2 [75]**	**Na_3Bi [76]**	**Stanene [77]**	**Stanene Tunnel [78]**
L_G (nm)	10.32	10	10	1000	2.7	14.98	16.9
$Width$ (nm)	7.44	48	48	45	7		
t_{ox} (nm)	10	18	14				
V_{DS} (V)	0.05	0.65	0.65		0.05	0.05	0.2
$V_{TG} - V_{BG}$ (V)	1.84 V (ON)				11 V (ON)		
$V_{TG} - V_{BG}$ (V)	3.1578 V (OFF)				20 V (OFF)	10 V (OFF)	
I_{ON} (μA)	12.452	41.952	37.776				
I_{OFF} (nA)	0.9882	0.48	0.48				
I_{ON} ($\mu A/\mu m$)	1673.655	874	787				
I_{OFF} ($nA/\mu m$)	132.82	10	10				
I_{ON}/I_{OFF}	1.26×10^4	8.74×10^4	7.8687×10^4		6		4.9×10^2
SS (mV/decade)	119.01	82	72	305			
C (fF/μm)	0.1686	0.44	0.39	0.19			
τ_d (ps)	0.0069	1.06	0.96	.19			
DED (fJ/SE)	4.3448×10^{-6}						
DPD (pW)	2.1914	0.65	0.49				
SPD (pW)	0.004941						

In Table 5.2, the device length is approximately equal to the IRDS 2025 standard, about 10.32 nm, although the device width is comparatively smaller than the IRDS standard. The Drain-to-Source voltage for our device is 0.05 V, which is much smaller than the IRDS standard. According to the IRDS Intel roadmap, an ideal subthreshold swing of 60 mV/dec and a current on-off ratio of 10^5 are critical for optimal device performance. To meet these benchmarks, a Drain-to-Source voltage (V_{ds}) of at least 300 mV is generally required. Operating below this voltage such as with a V_{ds} of 50 mV could result in suboptimal performance, with limited current modulation and a reduced on-off current ratio, which could degrade the device's characteristics. While low-voltage operation has advantages like improved noise immunity, maintaining a V_{ds} of at least 300 mV is essential to meet IRDS standards and ensure the desired

switching behavior and current ratio for practical applications. We know that thermal voltage

$$V_t = \frac{kT}{q}$$

is 26 mV at 300 K. At this temperature, noise immunity is unstable because the Drain-to-Source voltage (V_{ds}) of 50 mV is close to the thermal voltage. Here, noise immunity refers to the device's ability to resist the effects of electrical noise. FETs are particularly sensitive to noise at low voltages and currents because the current passing through them is very small. A low V_{ds} means the device operates near the threshold voltage, where small voltage fluctuations can significantly affect the current flow and the device's operation. By operating at low temperatures, thermal noise decreases, thereby improving noise immunity. At 77 K and 4 K, the thermal voltage will be 6.35 mV and 0.368 mV, respectively. Therefore, since thermal noise is directly related to temperature, lowering the temperature reduces the noise, which is beneficial for low-voltage operation.

Table 5.3: Comparison between High-k and Low-k Effect and Performance Analysis on TFET

Gate Oxide	ε_{ox}	t_{ox}	τ_d	**DED**	**DPD**	**SPD**	**EOT**
unit	**(pS)**	(nm)	**(fJ/SE)**	**(pW)**	**(pW)**	**(pW)**	(nm)
HfO_2	25	14	0.0069	4.344×10^{-6}	2.1914	0.00494	1.932
h-BN	3.29	14	0.00839	5.225×10^{-6}	2.6355	0.01441	14.51

The comparison in Table 5.3 demonstrates that HfO_2 offers several advantages over h-BN for TFETs. HfO_2 shows lower dynamic energy dissipation (DED) at 4.344×10^{-6} pW compared to h-BN's 5.225×10^{-6} pW. Additionally, it outperforms h-BN in dynamic power dissipation (DPD), with HfO_2 at 2.1914×10^{-6} pW versus h-BN's 2.6355×10^{-6} pW. Static power dissipation (SPD) is also significantly lower in HfO_2 (1.00494×10^{-6} pW) compared to h-BN (1.441×10^{-5} pW). The Equivalent Oxide Thickness (EOT) is a crucial parameter in determining the performance of gate dielectrics in transistors. It is defined by the following equation:

$$EOT = \frac{\epsilon_{\text{ox}} \cdot t_{\text{ox}}}{\epsilon_{\text{SiO}_2}}$$

where ϵ_{ox} is the permittivity of the oxide material, t_{ox} is the physical thickness of the oxide layer, and $\epsilon_{\epsilon_{\text{WTe}_2}}$ is the permittivity of the material used for the gate dielectric, such as HfO_2.

A higher EOT typically indicates improved leakage current control and better device reliability, although it may slightly reduce electrostatic control. This trade-off is critical for optimizing power efficiency and long-term stability in TFET devices, where the choice of dielectric material plays a significant role in overall device performance. Here, HfO_2 has a higher equivalent oxide thickness (EOT) of 12.5 Å, compared to h-BN's 0.5214 Å, which improves leakage current control and device reliability. While the higher EOT may slightly reduce electrostatic

control, it brings substantial benefits in terms of reducing power consumption and enhancing long-term stability. Overall, HfO_2 is more power-efficient and reliable than h-BN, making it a better choice for TFETs, especially in low-power, high-performance applications.

CHAPTER 6: CONCLUSIONS

6.1 Conclusions

In this thesis, we have thoroughly examined the transport and electrical properties of 1T′-WTe_2 monolayers and nanoribbons, emphasizing their topological features and possible device uses.

Initially, we looked into the 1T′-WTe_2 monolayer's electronic band structure and PDOS. By examining the evolution of WCCs, the topological structure of the system was verified and the band inversion caused by SOC was readily identified. Additionally, we investigated the topological phase transition caused by inversion symmetry breaking from the QSH phase to a trivial insulator phase. We discovered that this transition takes place at a critical electric field of E=0.1431 V/Å.

Next, we analyzed the electronic and transport properties of both armchair and zigzag 1T′-WTe_2 nanoribbons. A bandgap was found for 1T′-WTe_2 monolayer but we found no bandgap for 1T′-WTe_2 nanoribbons, resulting in edge states around the Fermi level. The application of an external electric field opens a band gap in both armchair and zigzag nanoribbons. We also investigated the robustness of edge states in the presence of vacancy defects in zigzag nanoribbons. While trivial edge states and bulk states are significantly affected by the defects, non-trivial edge states remain protected due to TRS.

Subsequently, we examined the transfer characteristics of our proposed TIFET. Our findings indicate that an increase in channel length resulted in a decrease in both the subthreshold slope (SS) and the OFF-current, which is ascribed to the attenuation of tunneling current during the OFF state in elongated channels. Furthermore, we assessed the efficacy of the dual-gate edge-passivated zigzag 1T′-WTe_2 nanoribbon TIFET. The proposed TIFET demonstrates advantageous performance for on/off ratio, off-current, capacitance, delay time, and dynamic energy dissipation. Furthermore, The subthreshold swing (SS) of the device is relatively high compared with the IRDS 2025 standard.

In general, this thesis offers insightful information about how topological characteristics and electronic device performance interact. Our research establishes the foundation for the advancement of next-generation, low-power, high-performance electronic devices utilizing topological materials.

6.2 Future Prospects of Our Work

The current studies in this area are directed at solving the limitations of performance by finding materials with a greater bulk bandgap of about 300mV that will enhance both on-current and off-current. A strategy that will be important in improving efficiency to bring the transistor in line with the standards of IRDS will be to lower the gate voltage to increase the subthreshold swing. It is also under investigation that bilayer, trilayer or multilayer configurations of 1T′-WTe_2 may be used to reduce the electric field needed to induce topological phase transitions. The next promising direction is a way that can regulate and reverse these transitions by controlling the electron concentration instead of using only an external electric field. In addition to the digital applications, the devices are also being considered in analog application and in this case, careful examination of particular measures such as conductance (gds) and transconductance (gm) is necessary. Also, there is a push towards CMOS-like functionality, using two-phase transitions, between topological and trivial phases, to replicate the behavior of PMOS and NMOS devices. Lastly, the low-defect material growth, as well as 1T′-WTe_2, which is used as a critical interconnection material, is the key to the future integration of high-performance digital systems. The process of optimizing these nanoscale devices for future applications is analogous to optimizing a high-performance engine; each of the manipulations on the fuel mix (electron concentration) or the cylinder design (material layers) is geared towards maximizing power (current) and yet making the system as cool and efficiently running as possible.

6.3 List of Publications

- J. Sharma Md. Niloy Khan, M. Alam (2024). "Topological phase transition in 1T-WTe 2 zigzag and armchair edge nanoribbons with the applied vertical electric field for application in FET," *Journal of Physics: Condensed Matter* , (Under review)
- J. Sharma and M. Alam, "First principle study of topological phase transition in 2D 1T WTe2 bulk monolayer with the application of vertical electric field,"

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Design And Performance Analysis Of A Low-Energy Topological Field Effect Transistor Based On The Phase Transition In 1T′-WTe_2 Nanoribbons

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Dec
2025